
M@JI-CS

M@JI Computer System

USER AND TRAINING MANUAL

M@JICS Version 3.7

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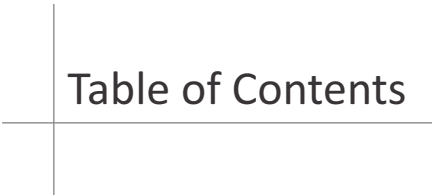
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1. General

1.1. SYSTEM OVERVIEW:

The system uses a standard Graphical User Interface (GUI) adopted from Microsoft Windows™ metaphor used for Menus and Icons.

The System is launched by double-clicking on the SULIS M@JICS CMIS icon (Normally found on the desktop or On the Start Menu in All Programs>sulis) when installed. The Login screen comes up with the Welcome screen in the background. The user will enter the assigned Username ID and the Password to login. In other cases, the user is required to select the name of the database they want to access from a list. The system will authenticate the user name and password then authorize the starting of the program, which will open the software desktop with the menus loaded. (Refer to figure 3)

The menus available to the user will depend on the type (role) of user.



Figure 1: Software Icon



Figure 2: M@jics Welcome Screen



Figure 3: M@jics Login Screen

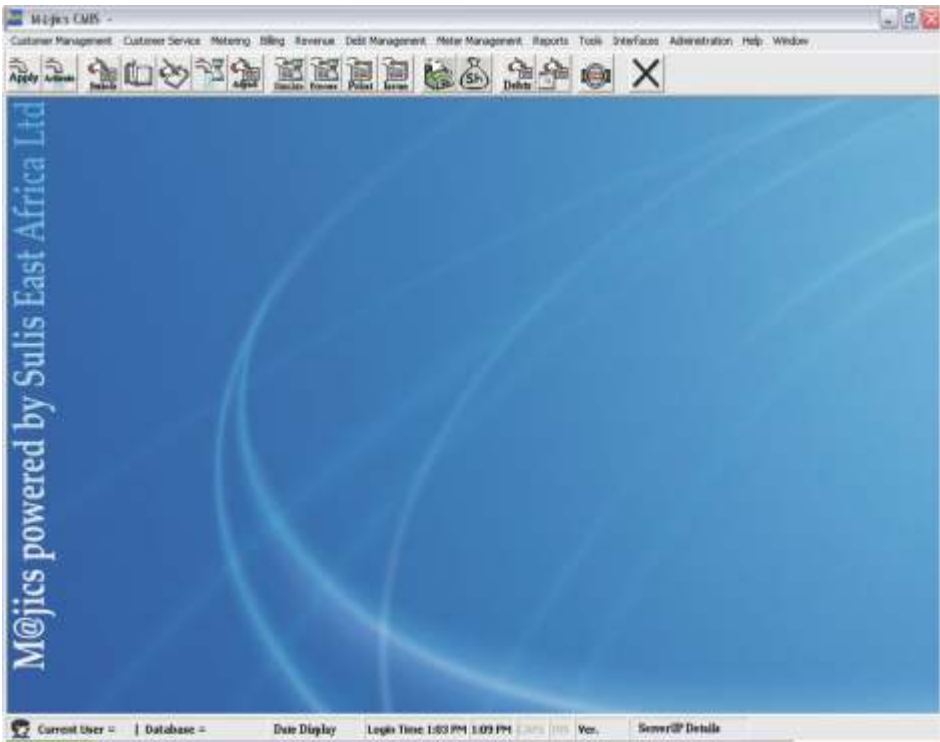


Figure 4: M@jics Desktop Screen

2. | Customer Management

2.1. CUSTOMER MANAGEMENT GENERAL DESCRIPTION

The function has a set of sub functions that facilitate customer management. They include the capture, editing and updating of customer details, the activation and termination of contracts and the display of contract-related (customer, connection) information.

On selecting the Customer Management menu, the following possibilities (menu items) exist:



Figure 5: Customer Management Menu

2.2. APPLICATION FOR NEW CONNECTION

This function facilitates the entry of customer and contract details from an application form of a new customer. This capture is done into the system after a customer fills out the application form, the survey is done and some /all payment is done.

Figure 6: Create Application Entry Form

Precautions:

The following fields **MUST** be entered before the application can be captured into the system:

- ☐ Type of Customer, Customer Name and Landlord Name if any, Application Date, Account Number, National ID, PIN No, e-mail, Billing Postal Address
- ☐ Select the Area, Billing group and Sub-zone; customer typology
Select if the customer is connected on a sewer line.

On filling all the required information, click on Create Application button and the application of the account is created in the system.
The status of the account application is set to Pending.

To continue with the application process,

- The user should ensure deposits and payments for the account have been captured in the system
- And a meter has been installed to the customer's account in the system.

To know the procedure for activating the customer's account (See Chapter on Activating an Account.)

2.3. SUPERSEDE ACCOUNT/CONNECTION

This function allows the user to assign an existing connection to a new user i.e. to create a new contract for an existing connection. The connection must be existing on the ground (the connection is available or there exists a meter) and the older account of the connection inactive. This function can only be invoked for an existing inactive/terminated connection. The function displays a screen to help the user search and display the older connection and assign it to the new contract.

Layout:

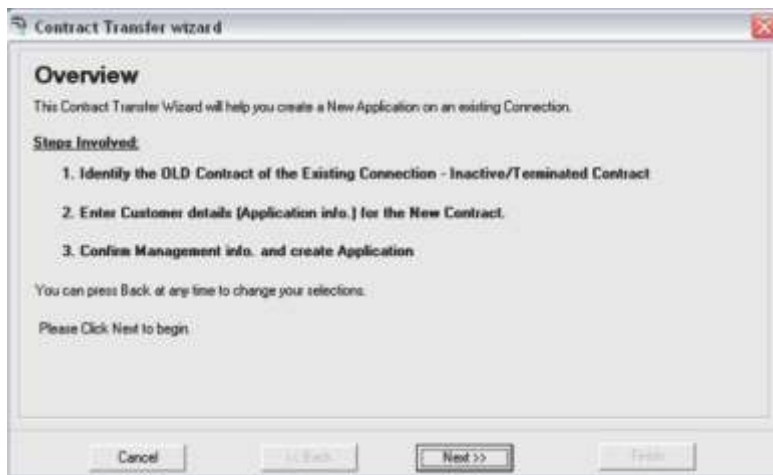


Figure 7 : Supersede Account/Change of User Wizard

On selecting the Next button, you will be taken to the next screen which involves selecting the old contract.

- Use the Search Button to pick out the search criteria for identifying the old contract in question. (Refer to Chapter on Search Screen)
The details will be displayed in the window
- Click Next to move to step 2 of 3 which will require you to select the new pending application. Enter the required fields to complete the details about the pending application.
- Click Next to move to step 3 of 3 which will allow you to enter the application details (eg. Customer Route, Typology and Category) or leave the existing details.
- Click Finish to complete the process.

2.4. DUPLICATE APPLICATION

This aids the MIS user create new applications using an existing contract.

- An account no, connection no. and date of application is needed
- Information of the selected customer are loaded and can be edited.

This feature is useful where a customer has more than one account or connection with similar details that can be used to create another application.

The screenshot shows a software window titled "Duplicate Application". It contains two tabs: "Customer profile" and "Contract Details". The "Customer profile" tab is selected and displays various input fields. On the left, there's a "Type of Customer" dropdown menu. Below it, the "Customer" section includes "First Name" and "Other Names" text boxes. The "Land/Lord" section includes "First Name" and "Other Names" text boxes. The "Billing Postal Address" section includes "P. O. Box", "Town", "Post No.", "Street Loc", and "Zip code" text boxes. Below this is a "National ID" text box and an "Employee Certificate" text box. On the right side of the "Customer profile" tab, there are fields for "Account No.", "Connection No.", "Application Date" (with a calendar icon showing "11 Sep 2010"), "Zone" (dropdown), and "Area Office" (dropdown). The "Contract Details" tab is partially visible behind the first. At the bottom of the window are three buttons: "Create Application", "Search", and "Close".

Figure 8: Duplicate Application Entry Screen

- Click the search button and obtain the required customer account to duplicate.
- In the Customer Profile, Enter the New Account No., the Connection No, and Application Date.
- Change the required fields, for example; change the Customer Typology to Water Kiosk, Customer Category and Disconnection Profile.
- Once you are done with any updates of the new application, click the Create application button to complete the process of duplicating the application.

2.5. MODIFY APPLICATION

This is a function used to change the customer profile details and contract details while the status of the customer is “Pending” i.e. before the contract is activated. The existing customer details in the application are displayed with the possibility of replacing some content in the provided fields found in the customer profile tab and the contract details tab. The CMS user will search for the customer and the details will be displayed for editing. (Refer to Chapter on Customer Search Engine to learn more about searching and finding optimum results)

The screenshot shows a web application window titled "Modify Application". It features two tabs: "Customer profile" and "Contract Details". The "Customer profile" tab is currently selected. The form contains several input fields and a dropdown menu. On the left side, there is a section for "Billing Postal Address" with fields for "P. O. Box", "Tower", "Hse No/E state", "Plot No", "Email", and "Telephone". Below this is a field for "Employer Certificate" and a "Comments on Customer" section. On the right side, there is a field for "Application Date" set to "25 Mar 2014", and fields for "Connection No", "Account No", "National ID", and "PIN". At the bottom of the window, there are three buttons: "Modify Application", "Search", and "Cancel".

Figure 9: Modify Application Screen

- Click the search button and select the customer/application to change.
- Edit the necessary details on each tab (Customer Profile, Contract Details and Connection Details).
- Click the Apply button to save the changes or click Cancel Changes to ignore any changes made.

2.6. CREATE WORK ORDERS

This is an organization's internal module that is used to compile a list of pending works with the help of the Information System. This allows the user to assign tasks for new or existing customers to a technician.

Figure 10: Work Order Capture Screen

- Select the range of dates that the applications were made.
- Select the task type & billing zone or “All” option to display all applications received between the date range.
- Click the Load Accounts button and a list will appear at the unallocated accounts section. Select Accounts by clicking the check box to tick.
- Next step is to allocate Technician concerned by selecting the drop down menu. Type in the name then Click the Add button to create a new technician.
- Click the Save Order button to create the work order.
- Click the Print Order to produce a hard copy that will contain details of the order.

2.7. UPDATE WORK ORDERS

This function facilitates the alteration of the statuses of the work orders. The user can add or change remarks or status, cancel and reactivate tasks.

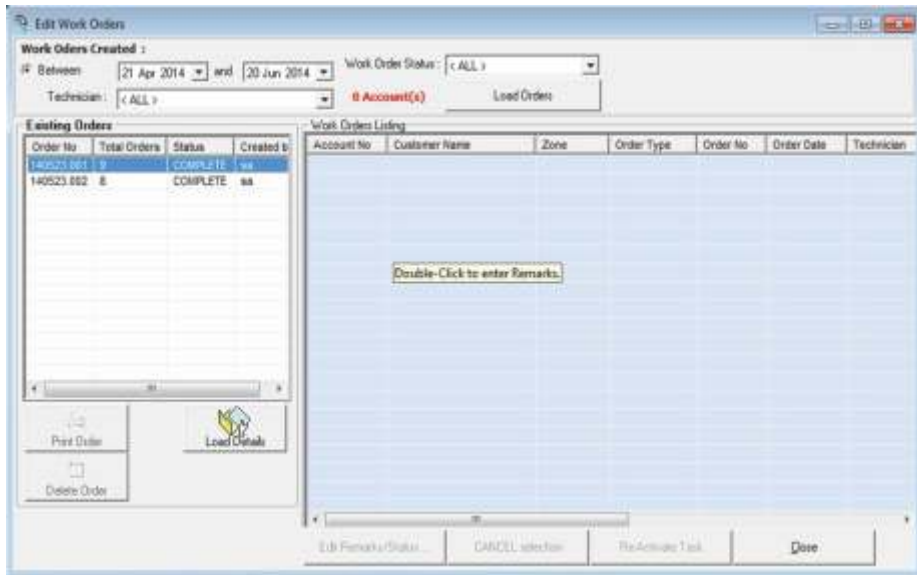


Figure 11: Edit Work Order Capture Screen

- Select the range of dates that the work orders were created.
- Select the required technician or “All” option to display all work orders assigned to all technicians.
- Select the work order status to filter the work orders required
- Click the Load List button to view work orders. Click on the Load details button to view more information about an existing order.
- Select order(s) by clicking the check box to tick after the list is loaded.
- At the bottom of the screen select the button that is required to achieve the desired task, i.e. re-activate tasks or change status
- Click the Close button to exit the screen.

2.8. WORK ORDER REPORTS

The sub menu includes the following reports

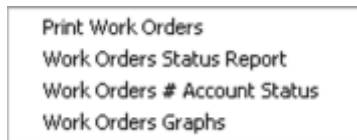


Figure 12: Work Order Reports

2.8.1. *Print Work Orders*

This screen allows one to choose the period that the work orders were created. It also provides for selection of the technician.

The following is the layout:

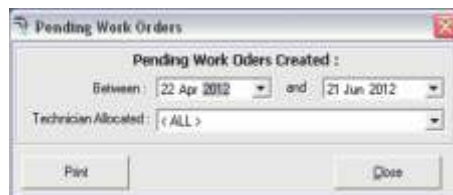


Figure 13: Print Work Order Screen

- Select the range of dates appropriate when the work orders were created.
- Select the name of the technician or All from the drop down menu to produce a report inclusive of all the technician's tasks.
- Click the print icon button to preview the report or to print to a printer. Clicking Close button exits the report builder.

2.8.2. *Work Order Status Report*

This screen allows one to create a report per work order status.

It also allows for selection of the technician assigned.

The following is the layout:



Figure 14: Print Work Order Per Status Screen

2.8.3. *Work Orders Vs Account Status*

This provides various reports, these include



Figure 15: Work orders Vs Account Status Report Builder

- Select the desired report and select Print button to view
- Pending Application without work orders report provides data on all applications that have not been assigned.
- List of all pending applications with orders that have been carried out.
- Analysis of all pending work orders with accounts that are active

2.8.4. *Work Order Graphs*

This screen allows one to print out visualization of work orders.

It provides for selection of the period, technician allocated and work order status.

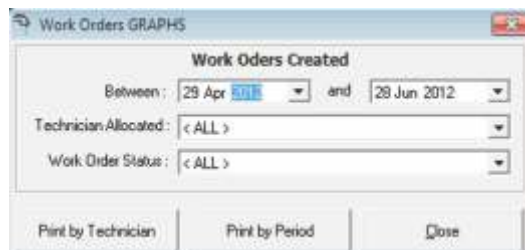


Figure 16: Work Orders Graphs Builder

2.9. ACTIVATE CONTRACT

This submenu changes the status of the customer/connection from a pending state to an active state allowing the client to receive services from the company.

Note:

It is possible to activate a contract if:

- a) The full deposit is paid
- b) Any new connection charges are paid
- c) All the customer, Contract and connection details have been captured. (Customer personal details, customer management information, contract details, connection payments, connection and metering information).

See layout below:

Figure 17: Activate Contract Screen

- Select the activate connection from the customer management menu.
- Click the search button to obtain the pending connection or customer to activate.
- Enter the necessary connection details such as meter type, number, connection size and length by installing the meter.
- After confirmation that all the required fields are entered, click the activate Connection Button.

2.10. MODIFY ACCOUNT DETAILS

This is a function used to change the customer profile details and contract details while the status of the customer is “Active” i.e. After a contract has been activated. The existing customer details in the application are displayed with the possibility of replacing some content in the provided fields found in the customer profile tab and the contract details tab. The CMS user will search for the customer and the details will be displayed for editing. (Refer to Chapter on Customer Search Engine to learn more about searching and finding optimum results)

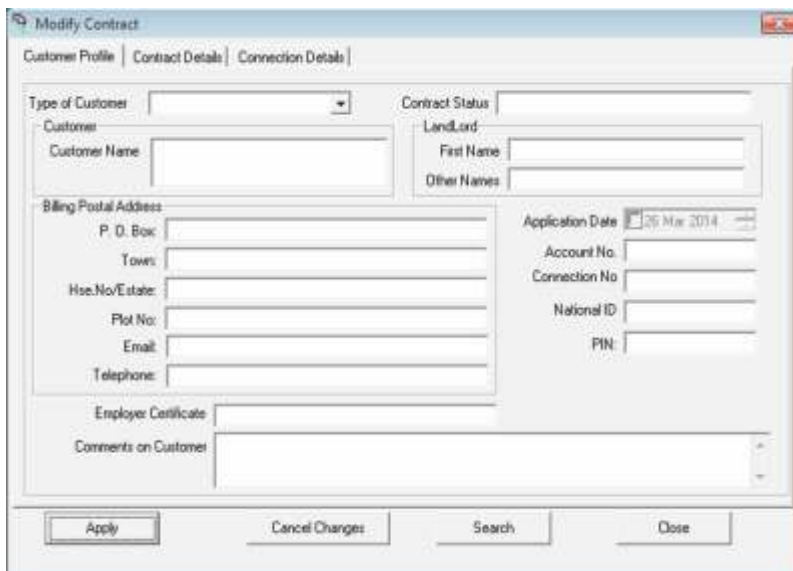


Figure 18: Modify Application Screen

- Click the search button and select the customer/application to change.
- Edit the necessary details on each tab (Customer Profile, Contract Details and Connection Details).
- Click the Apply button to save the changes or click Cancel Changes to ignore any changes made.

2.11. DISPLAY ACCOUNT DETAILS

The function enables CMS user to view the details of a customer, contract details, the connection, the Metering, Billing, Payment details, Events, Disconnection History, Meter Movements, other information and Reading details. The details are available as a report which can be printed. This is one of the most frequented screens in the program.

The layout is as below:

The screenshot shows a software window titled "Display Contract Details". It features a tabbed interface with tabs for Billing Details, Payments Details, Events, Disconnection History, Meter Movements, Customer Profile, Contract Details, Connection Details, Other Information, and Readings Details. The Customer Profile tab is active, displaying a form with fields for Account No., Customer ID, Customer Name, Billing Postal Address (P.O. Box, Town, P.O. Code, P.O. No., Email, Telephone), Employee Certificate, Comments on Customer, Connection Status, Type, Land/Lord First Name, Other Names, Date, Account No., Connection No., National ID, and PIN. At the bottom, there are buttons for Search, Print, Print Statement, Email Statement, Historical Statement, and Close.

Figure 19: Display Account Details Screen

An example of customer details....

The screenshot shows the same "Display Contract Details" window, but with data loaded. The Account No. is 03123-001, Customer ID is 14010009408, and Customer Name is UNISPAN LIMITED. The Billing Postal Address includes P.O. Box 31377-00600, Town NAROBI, P.O. Code APPROVE, P.O. No., Email, and Telephone 0723621769. The Connection Status is Terminated Connection, Type is Individual, and Land/Lord First Name is UNISPAN LIMITED. The Date is 07 Jul 2010. The Account No. is 03123-001, Connection No. is 03123, National ID is 0, and PIN is. At the bottom, there are buttons for Search, Print, Print Statement, Email Statement, Historical Statement, and Close. The status bar at the bottom indicates "Last updated By SA on 02 Mar 2014 12:54:45 Status Terminated Connection".

Figure 20: Loaded Account Details of a Customer

An example of a print report...

Customer Summary Report	
Customer Profile	
First Name : DANIEL OHALLO MUSIT	Last Name : DANIEL OHALLO MUSIT
Other Name :	Last Name (Old) Name :
Account No. : 400110	Business Contract :
Activation Date : 13-Oct-2006	Area Code :
Interval ID : 0	KRM Hakamaga
PR :	Zone :
Address (Full Address) :	KRM Hakamaga
P.O. Box : BOX 1602 KAKAMEGA	Connection Contract :
Town : KAKAMEGA	
PERIOD : NA	
Days Off : NA	
Bank :	
Telephone :	
Contract Details	
Contract No. : 400110	Deposit Amount : 0.00
Contract Date : 13-Oct-2006	Deposit Month(s) :
Contract No. : 400110	Deposit Date :
Management Information	
Bill Type :	Contract Package :
ANAL AMLEMDA	DOMEST DOMESTIC
Contract Category :	Disconnection Type :
DST DOMESTIC	NOR Normal Disconnection
Code :	Average Disconnection : 0.00 RD
Connection Date :	The Activation Date(s) : 13-Jul-2010
Connection Date : 13-Oct-2006	
Accountance :	
	1,445.00 Customer's Debt
Connection Details	

Figure 21: Customer Summary Report

For the Reading Details, Billing Details, Payment Details, Events, Disconnection History, and Meter Movement tabs:

- Click the print button to view a printable view of the details.

The **Print statement** button prepares a list of all the transactions (credit and debits) an existing customer has had since they made an application for a connection/account.

The **E-mail statement** button will enable the CMS user to send a statement via e-mail. This will only be applicable when the customer details have an e-mail address account of the customer.

The **Historical statement** button will enable the CMS user to view a statement that has data from Older systems that have been replaced. This will be available for reference purposes only.



Figure 22: E-mail Customer Statement Screen

The e-mail screen contains fields for entering information such as e-mail address, the subject attachments and any accompanying text for the customer.

When you click the send button, the email and its attachment will be sent to the addressed picked from the customer account details.

Note: For you to send emails to customers, the email address must be captured in the customer account details.

2.12. INACTIVATE CONTRACT

This function allows an “Active” contract to be made “dormant” such that no meter reading or billing can be done on the connection. Payments can still be captured on an inactive account and refunds are possible for these accounts.

Requirement:

To inactivate a consumer, they must be an existing active client of the company as well as in the system.

Reasons for inactivating:

- If a customer moves out of a residential house and requests that the company deactivates the account.
- If a customer has not paid bills for a certain period of time which is determined by the company policy.

Note:

Once a contract is deactivated, a connection can be made active by reactivating it. It can be reactivated or terminated.

Layout:

Figure 23: Inactivate Contract Screen

To inactivate a contract:

- Click the search button to obtain the required contract or customer account to inactivate.
- The details of the customer will be displayed on the form to confirm the selection.
- Click the inactivate button to make the contract dormant.

2.13. REACTIVATE CONTRACT

This function allows an “Inactive” or dormant account to be reactivated. This will allow the normal transactions and services to be resumed.

Reason for Reactivating:

- A reactivation would occur if a customer would have returned to the same premises after he/she had requested the contract be inactivated while he/she was away.
- If any new customer wants to start using a dormant connection.

Requirement:

- The contract should have been inactive before the process is begun.
- The contract can only be reactivated if the customer has no unpaid debts (i.e. all the bills have been cleared).
- The associated Billing Period is open.

The following is the Reactivate Contract layout:

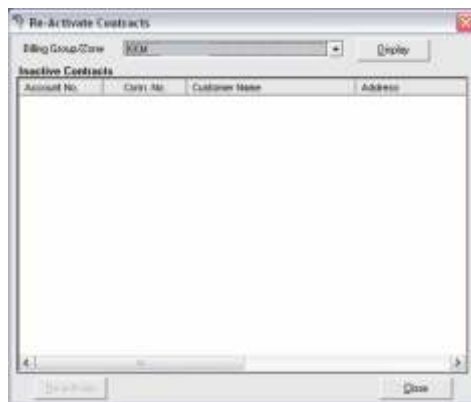


Figure 24: Reactivate Contract Screen

To reactivate a contract:

- Select the Billing Group or zone that is affected using the drop down menu
- Select the dates in which the contracts were inactivated. Specify the start and end date.
- Click the Display button to show the inactive contracts.
- Select the contract that you are interested in activating.
- Click the Reactivate button to complete the process.

2.14. TERMINATE CONTRACT

This function allows an “Inactive” Connection to be closed completely. This is done when a customer has left the premises or when the customer has indicated that he/she will not require the services of the company and requests that the connection be closed.

Requirement:

- The contract can only be terminated if the customer has no debt (i.e. all the bills have been cleared).
- The deposit has been refunded to the consumer or used to clear the debt where applicable.

The following is the layout:

Figure 25: Terminate Contract Screen

To terminate a contract:

- Click the search button to obtain the required inactive connection
- The termination comment area will display information relating to the status of the contract/connection stating if it is allowable to terminate or not

- Depending on the information displayed on the termination comments section choose to terminate the contract by clicking the Terminate button or click the **Update deposit** button to correctly bring up to date the status of the customer account with deposit refunds.

The following layout is an example of an account ready to be terminated:

Terminate Contract

Account No. 100502 of E. LEEN MALOVA

Customer Profile | Connection | Other Information | Terminate Contract -

Terminate Contract -

Reference : Ledger No

Terminate Date :

Termination Comments

This Connection has a NIL Balance. Termination is Possible

Search Close

Last updated By SA on 13 Jul 2010 12:33:56 PM Status In-Active Application

Figure 26: Terminate Contract Screen showing Balance

- Enter the reference number and a ledger number where applicable. Select the date that the contact was terminated.
- Click on the **Terminate** button to update the status of the account
- Click the **CLOSE** button to exit the process or exit the window.

Note: It is possible to use the deposit to settle an account's debt on condition that the amount is equal to or less than the deposit amount.

2.15. DISCONNECTION REQUEST

This function is used to capture details of customers who would like to have their connection disconnected for a period due to their unavailability on the premises.

According to a company's policy, a connection can be disconnected due to the following reasons

- The tenant or owner of an account requesting for disconnection after making a payment of the associated charge.
- The customer having overdue debts.

The following is the layout:

Figure 27 : Disconnecting a Connection Screen

To disconnect a connection:

- Click the search button to obtain the account to disconnect.
- The details of the customer will be displayed in the form together with the connection and other information details. In the Disconnect Connection tab, enter the disconnection date, reading obtained, the disconnection mode and the name of the artisan conducting the disconnection.
- Click the **Disconnect** button to complete the process.

2.16. RECONNECTION REQUEST

This function is used to capture details of a consumer who has had their connection disconnected for a period of time.

According to a company's policy, a connection can be reconnected when

- The customer has paid all overdue debts and all required top ups and charges are made in the system.
- The customer resumes to using the connection after returning to his/her premises.

The following is the layout:

Account No.	Conn. No.	Customer Name	Disconnection Date
-------------	-----------	---------------	--------------------

Figure 28: Reconnecting Connections Screen

To reconnect a connection:

- Select the Billing Group or zone that the customer is situated in.
- Select the range of the dates when the disconnections were made (Start and end date).
- Click the **Display** button to show the disconnections made between the selected period.
- Scroll from the list and find the connection required, select it and click the **Reconnect** button.

2.17. REPORTS

This is a function that is used to generate a summary of the customers, the contracts and connections that the company has had for a stated period of time. The following is a sub menu of the reports found in the Customer management menu.

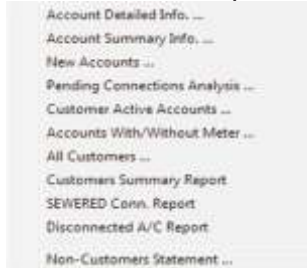


Figure 29: Customer Management Report Menu

2.17.1. *Account Detailed Info*

This menu allows one to select a single customer account to view information about their contract. It gives information pertaining to a customer in an exhaustive manner. The report includes information such as Customer Profile, Contract Details, Management Information, Connection Details, Metering information, Billing Summary details, Payment summary details, Meter Reading, Meter Movements, Connection History, Complaint History and Events. The following is the layout:



Figure 30: Contract Detailed Information Report Builder

- Obtain the customer account number from the Customer Search Engine.
- Select the report range in terms of the start date and the end date. This will apply to the Metering info, Billing, payment summary, meter reading, connection history complaints history and Events.
- Click the print icon button to preview the report or to print to a printer.

2.17.2. Account Summary Info

This report builder allows you to select a customer account that you would like to view the contract details for. It gives an outline of information, pertaining to a customer. The report includes information such as Profile, Customer Contract Details, Management Information, Connection Details and Metering information.

The following is the layout:



Figure 31: Account Summary Information Report Builder

- Use the search button to obtain the customer account number
- Click the print icon button to preview the report or to print to a printer.

2.17.3. New Accounts

This report builder allows you to generate a list of all new accounts created under a period range and found in a zone. The report will have information such as Account Numbers, connection info and meter details average consumption and deposits paid.

The following is the layout:



Figure 32: New Account Report Builder

- Select the office and the zone you would like to view the information (Select <All> to view information from all Billing groups).
- Select the report range which will include the Start and End date.
- Click the print icon button to preview the report or to print to a printer.

2.17.4. Pending Connection Analysis

This report builder allows you to produce a list of all awaiting contracts that have not yet been activated. The report produces all customer contracts that have not been activated and orders them by zones.

The following is the layout:



Figure 33: Pending Applications Report Builder

- Select the scheme name which will include all the zones.
- Click the print icon button to preview the report or to print to a printer.

2.17.5. Customer Active Contracts

This report builder allows you to generate a list of all contracts that have been operational. The report shows all the customer contracts that have active connections and disconnected connections. It provides information such as the customer name, Account number, connection no, Meter Number, average consumption and if the connection is connected to the sewer.

The Print summary report gives statistics about the billing groups which includes: group names, number of customers and number of sewered customers.

The following is the layout:



Figure 34: Active Contracts Report Builder

- Select the office and the billing group that you would like to produce the report for.
- Click the Print icon button to preview or click To Printer option to send report to printer directly.
- Click the print Summary button to view the billing group summary information as indicated above.

2.17.6. *Connection with/without Meter*

This report builder allows you to produce a list of all connections that have meters physically installed on site or those with no meters installed. As a default, a report without meters is produced. The report contains details such as account no, connection no, connection dates and billing addresses. There are two options in viewing the reports obtained; viewing the report with connections that have no meters or viewing connections that have meters. The following is the layout:

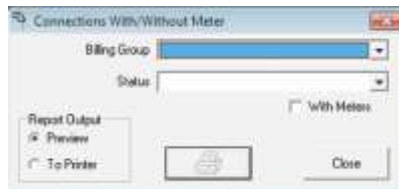


Figure 35: Connections with / without Meter(s) Report Builder

- Select a Billing Group and the status of the connections that you would like to view.
- To view connections with meters, check (✓) the option “With Meters”. To view connections without meters uncheck the option.
- Click the print icon button to preview the report or to print to a printer.

2.17.7. *All Customers*

This report builder allows you to produce a list of all customers from a zone or all zones. By selecting the option between ordering by contract, by connection or the last account bearer on a connection, you have the power to determine the customer list to view.

The layout:



Figure 36: All Customers Report Builder

- Select the billing group and select between the provided criteria.
- Click the print button to preview the report or to print to a printer.

2.17.8. *Customers Summary Report*

This report builder allows viewing of the total number of accounts using various criteria. Select one of the options between printing by Zones, customer category or customer typology.

The layout:

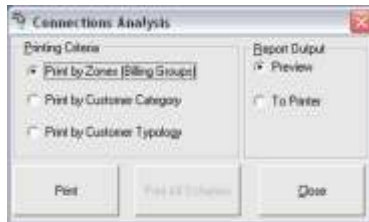


Figure 37: Customers Summary Report Builder

- Select the option that suites your needs.
- Click the print button to preview the report or to print to a printer, select to printer option on the right hand side.

2.17.9. *Sewered Connection Report*

This report builder allows viewing of accounts that are connected to sewerage and do not receive water from the water service provider due to various reasons.

The layout:

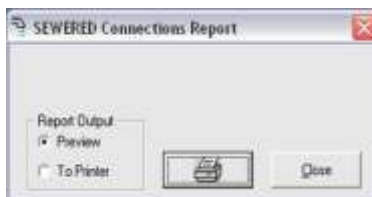


Figure 38: Sewered only Customers Report Builder

- Click the print button to preview the report or to print to a printer, select, to printer option on the right hand side.

2.17.10. *Disconnected Accounts Report*

This report builder allows viewing of the list of all the disconnected accounts per billing group. Select All Accounts or the zone concerned

The layout:

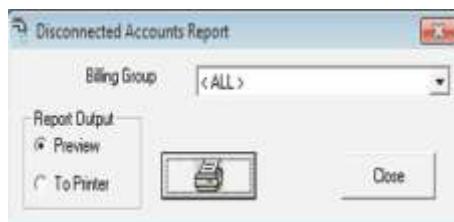


Figure 39: Disconnected Accounts Report Builder

- Select the billing group or zone required.
- Click the print button to preview the report or to print to a printer, select to printer option on the left hand side.

2.17.11. *Non Customers Statement Report*

This report builder allows viewing of statements of customers who do not have contracts with the company but have interacted with the company by obtaining certain services e.g. Exhauster services Dumping service, Tanker services, Tender sale etc

The layout:

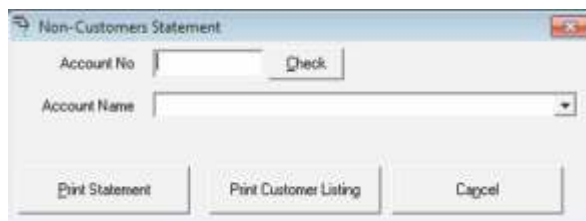


Figure 40: Non Customer Statement Report Builder

- Enter the Account No of the non customer or select the Account name from the drop down list then select Print statement.
- Click the print customer listing button to preview the report containing all the non customer names served by the company.

3. | CUSTOMER SERVICE

3.1. CUSTOMER SERVICE OVERVIEW

The customer service module is used to cater for complaints made by consumers. This module is key in helping the organisation optimize service delivery and customer satisfaction.

It will provide a platform for the organisation to record and keep track of any complaints brought in by customers as long as it has been captured in the module. The Menu consists of the following items:



Figure 41: Customer Service Menu

3.2. MANAGE COMPLAINTS

This feature is helpful for administering and managing all the complaints received from customer service assistants who interact with customers. The module displays unresolved complaints using criteria such as time (dates or days) and type of complaint categories found in the MIS, hence it has been coined as the INBOX.

Layout example:

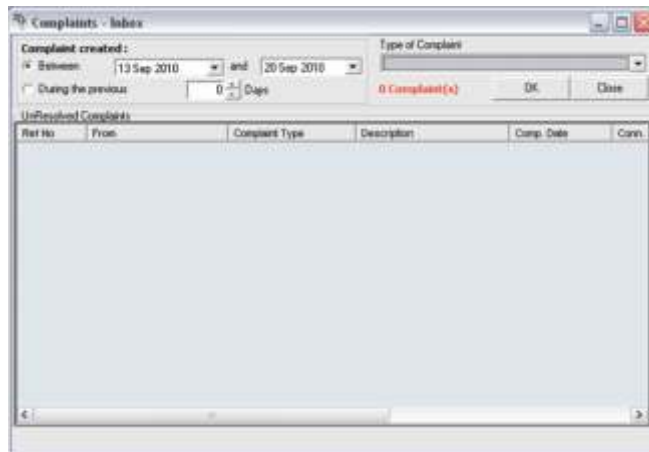


Figure 42: Complaints Inbox Screen

From this screen you can:

- Search for all the complaints made under a period by selecting the complaint type as “All” and the date range when the complaints were raised.
- Search for specific categories of complaints by selecting a type of complaint from the list and selecting a period. Click the OK button to view the results
- Right click on the task area to get the following menu:

Register New Complaint	- Create a new customer complaint
Modify Complaint	- Change details of a customer complaint you create
Register Action	- Create a reply for a complaint
Delegate Activity	- Remove replies to complaints you captured
Show History	- View actions taken on a complaint since recorded
Print the List	- Print all the complaints received without details
Copy Ref. No	- Place a complaint's reference no. on the clipboard

Figure 43: Manage Complaint Menu

3.3. REGISTER COMPLAINTS

The module is used to capture complaint issues raised by customers. It has the facility to tie a complaint to a customer, assign complaints to the respectful department in the company in order for action to be taken. Once the complaint has been captured and assigned, it appears in the INBOX of the employee responsible for the associated task.

The requirement is that a customer must be in the MIS database as an active or pending account holder. Non customer complaints are also considered by having a special code to represent all non customers.

The following image is the screen for registering a new complaint:

The screenshot shows the 'Register Complaint' window with the following fields and sections:

- Account No.:** A search icon and an 'OK' button.
- Customer Name:** A text input field.
- States:** A dropdown menu.
- Billing Address:** A text input field.
- Walk Route:** A text input field.
- P. O. Box:** A text input field.
- Walk No.:** A text input field.
- Tower:** A text input field.
- Connection No.:** A text input field.
- Plot No.:** A text input field.
- Zone:** A text input field.
- Supply Loc.:** A text input field.
- Complaint Category:** A dropdown menu.
- Complaint Source:** A dropdown menu.
- Complaint Type:** A dropdown menu.
- Section To Take Action:** A section containing:
 - Name:** A dropdown menu.
 - Position:** A dropdown menu.
- Priority:** A dropdown menu.
- Description:** A large text area for describing the complaint.
- Book Serial No.:** A text input field.
- Buttons:** 'Register Complaint' and 'Close'.
- Footer:** 'CRA - Customer Relations Assistant (Username)'.

Figure 44: Registering Complaints Screen

- Click the search icon to obtain the customer's account no., click OK
- Click the complaint category drop-down and select appropriate.
- Select type of complaint associated with the category type.
- Give the complaint a priority. Provide a complaint source too.
- Select the Department to take action and the position of the person responsible.
- Type in description of the complaint to further clarify situation.
- Enter a Book serial Number if available.
- Click Register Complaint button to record the complaint and forward it to the responsible department/employee.

3.4. MODIFY COMPLAINTS

The module is used to change or edit details about a recorded concern made by a customer. This can only be done only if you are the creator of the customer complaint in the system. Additionally, a complaint can be modified, only if it has been created on the same day.

The layout:

The screenshot shows a 'Modify Complaint' dialog box. At the top, there's a 'Reference No' field with a search icon, an 'OK' button, and a 'see details...' link. Below this is a 'Customer Profile' section with fields for Customer Name, Billing Address, P. O. Box, Town, Plot No, Supply Loc, Status, Route No, Walk No, Account No, Conn. No, and Zone. The next section contains 'Complaint Category', 'Complaint Source', 'Complaint Type', 'Section To Take Action' (with Name and Position dropdowns), 'Priority', and a 'Description' text area. At the bottom are 'Modify', 'Delete and Add', and 'Close' buttons. The status bar at the very bottom reads 'CSS1 - Customer Service Assistant 1 (Name)'.

Figure 45: Modify Complaints Screen

- Click the search icon button to obtain complaints already made within the day, or enter the reference number and click OK.
- The details will be displayed in the form, showing the customer profile and the complaints made, not forgetting the department assigned to take action.
- Edit any of the fields presented on the form and click Modify button to update the changes to the complaint.

3.5. DELETE COMPLAINTS

This module is key in removing already existing complaints, that have been captured erroneously into the system. This process can be carried out only within the same day the complaint has been captured. In addition, the complaints can only be deleted by the user who created them.

Layout:

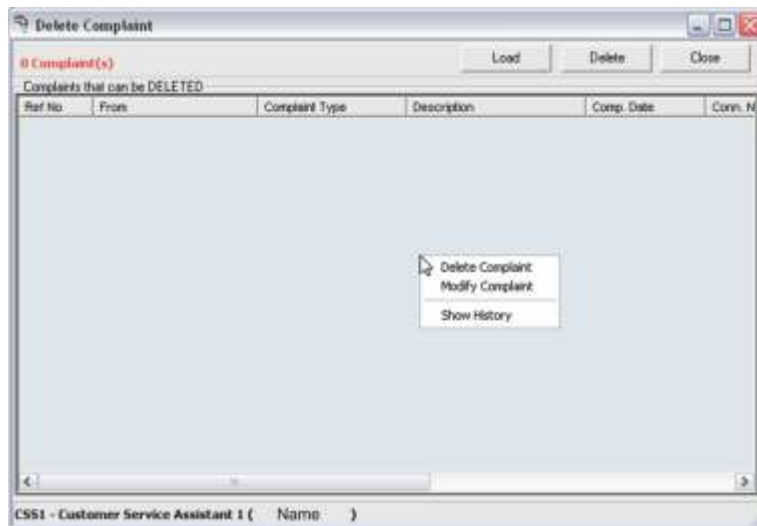


Figure 46: Delete Complaints Screen

- Click the Load button on the form to display the complaints that you have registered within the day.
- When the list has been displayed, select the complaint that was recorded inappropriately and click the Delete button.
- Click the Close button when you are done with the screen.

NB:

- Double click on the complaint that has been displayed to view additional information about the complaint.
- You can also right click on the complaint displayed to perform more actions such as Deleting the complaint, Modifying the complaint and showing the history of the complaints made.

3.6. REGISTER ACTION

The following module is used to register solutions or steps taken on a complaint that is existing in the MIS. In order to register an action, a complaint must have been assigned/sent to you, hence appearing in your INBOX.

You can delegate a received complaint to another employee in the same/ different department in the organisation giving a description.

Layout:

Figure 47: Register Action Screen

- Click the search icon button to obtain the complainant's reference number and click the OK button.
- The customer profile details will be displayed and summary information will be displayed. To see full details of the complaint, click "[see details...](#)" found at the top, next to the OK button.
- Next to that we have an Add comments button to be used to add progress of the complaint after being captured into the system.
- In the case of a resolved complaint, follow the above procedure enter the name of the section, position and your name as the employee taking action, this time enter details of the solved task in action description. Check (✓) the box labeled **complaint resolved**.
- Click the Register button to complete the process.

3.7. CUSTOMER COMPLAINTS

The module is used to display all complaints a customer has ever reported to the company. This is displayed as a list in the complaints history area, each row representing a single complaint registered.

The following is the layout:

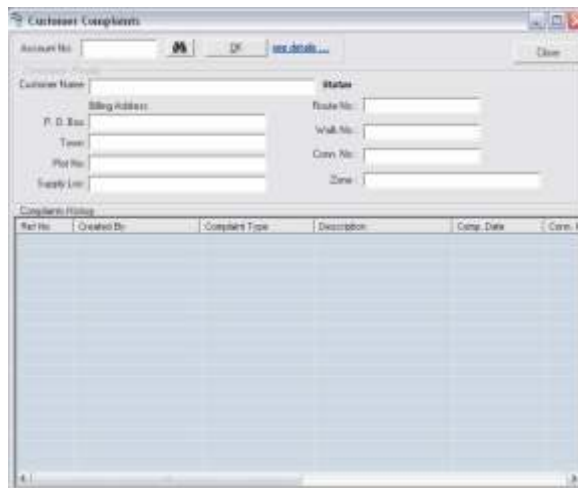


Figure 48: Customer's Complaints Screen

- Click the search icon button to obtain the account number of the customer, or enter the account number and click OK button.
- The customer profile will display basic details and the complaint history will display a list of all complaints a customer has made. This list will provide compact information about the complaints, to view details about each complaint, select it and double-click to open a detail view window. Right click to get the following menu.



- Return a complaint from a resolved state to unresolved state. To activate this feature, consult your administrator.
- Creates a new complaint for the loaded customer.
- Edits the complaint if you are the one who created it.
- Creates a task already completed for a complaint.
- Removes a task entered wrongly as an action.
- Displays a detailed view of the complaint recorded.

Figure 49: Customer's Complaints Menu

3.8. QUERY COMPLAINT

This is a module that enables a simple quick check of a complaint's progress for a customer who walks into the customer service desk or phones in to find out the progress.

The layout is as below:



Figure 50: Query Complaint Screen

- Enter the customers reference number they were provided with
or
Click the search icon to find the reference number using other search criteria.
Then
Click OK.

The following is a result of clicking the OK button on Figure 47.

Customer Profile

Customer Name: SARUL GOHING
P. O. Box: 410 KAWANGA
Town: KAWANGA
Plot No: 100
Supply Loc: 100

Complaint Details

Type of Complaint: Burst
Section to take Action: Technical Department
Position to take Action: TDS - Water Distribution Supervisor
Complaint Category: WATER DISTRIBUTION

Complaint Status

Priority: High priority
Status: Open - Unresolved Complaint
Resolved in: External - from Customer
Complaint Source: External - from Customer

Complaint Description

BURST PIPE THAT SHOULD BE REPAIRED AS SOON AS POSSIBLE

Complaint created by: mulya (CSSI - Customer Service Assistant 1) of Customer Service Department on 20-Sep-2010 12:29:43

History of Action

Action Type	From	To	Description
Complaint Registrati...	CSSI - Customer Servi...	TDS - Water Distri...	BURST PIPE THAT SHOUL...

Figure 51: Detailed Complaints Screen

- The first section, Customer Profile, will display summary details about the customer. The second section, Complaint Details, will display all the details about the complaint made and status of complaint. The third section is a history of all actions performed on the complaints. To view more details double-click on the action.
- To print the complaints click the Print icon button. Click the ok (✓) icon to close the window.

3.9. REPORTS

The module contains sub-items which hold screens with options to select features of the report to print or preview. Reports will provide analysis on various data from the MIS to provide crucial information about a service. The following is a list of available report for customer service:

- ☐ No. of Complaints
- ☐ Statistics & Graphics
- ☐ Un-resolved Complaints
- ☐ Complaints Register
- ☐ All Complaints

3.9.1 *Number of Complaints*

The report builder allows one to compile complaints collected from the customers according to the preferred parameters. The report will give a summary of all complaints received during a time limit.

The Layout:

Figure 51: Number of Complaints Builder Screen

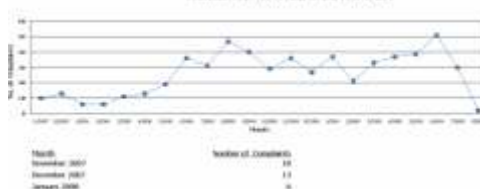


Figure 53: Monthly graphical report

- Select the date range you want view summary complaints on.
- Select the Zone, type of complaint, service source and service category.
- You can select the < All > option to include all the available parameters in the categories.
- Click the Detailed button to view complaint info. explained, Click the Graphical button to view statistical info with charts. As shown above in figure 49.

3.9.2 Un Resolved Complaints

This report provides a summary of all the complaints that are in a pending state. (To learn how to resolve complaints refer to chapter 3.6)

The layout is as follows:

- Select the required field with the exact information.
- Click the **Print** button to preview the report ordered by complaint category and department.
- To preview the report ordered by zones, click the **Print by Zone** button.

Figure 54: Unresolved Complaints Report Builder

3.9.3 All Complaints

This report puts together all information (statistics and Summaries) raised by the customer or raised internally by an employee for a customer. It provides statistics that can be helpful to the company in making concrete decisions in customer service.

The following is a layout:

Figure 55: All Complaints Report Builder

Select and choose the necessary options from the report builder to give results that are suitable for the information required. To view the information in a report , click the **Print** button, otherwise, click cancel.

3.9.4 *Statistics and Graphs*

This feature is available in the report section to allow the CMS user evaluate customer service complaints, tasks and solution in an empirical way using collected and organised data which can be represented graphically.

The layout is as follows:

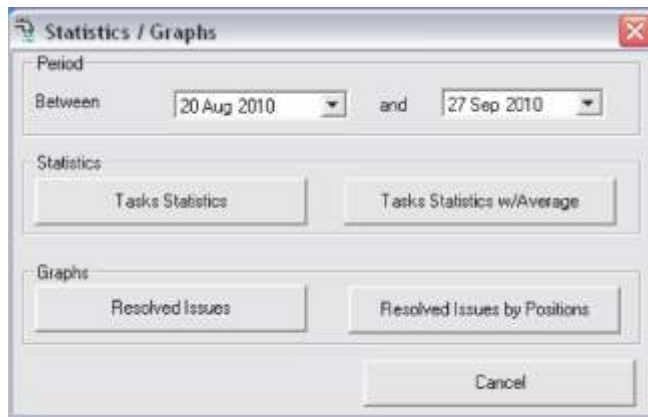


Figure 56: Statistics and Graph Builder screen.

- Select the date range/period you want view information about.
- If you would like to view statistics click **Tasks Statistics** which has the number of tasks or Task Statistics with Average which shows the average time taken to complete the tasks.
- To view graphical representation of the complaints collected, click the Resolved issues or Resolved Issues by Position button.

3.9.5 *Complaints Register*

This report is used to show all the complaints that were captured in the MIS.



Figure 57: Complaints/Issues Register

4. | Metering

4.1. METERING GENERAL DESCRIPTION

The following chapter describes the process of managing billing periods, reading books and meter readings collected. It includes features such as adjusting consumption, Route re-ordering, meter reader management, master meter management and group period statuses.

The following is a layout of the Metering menu:

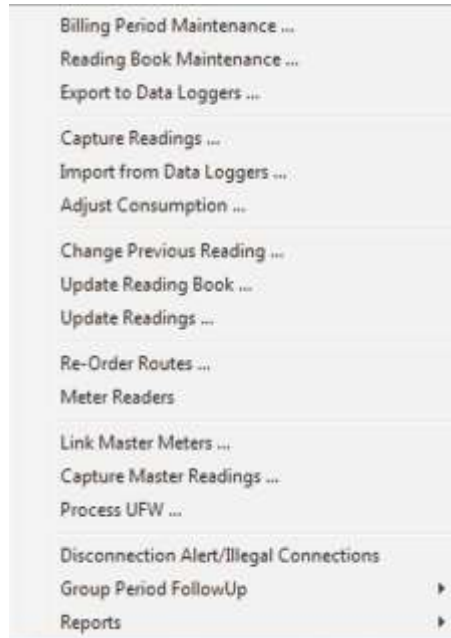


Figure 55: Metering Menu

4.2. BILLING PERIOD MAINTENANCE

The purpose for this feature is to allow management of billing periods for zones. The two options available are creating and deleting billing cycles. A period can only be deleted if no Meter readings have been captured. The user selects the billing group/zone and the billing period to create a cycle. A new period can only be created when bill processing is complete. The form will show details of billing group and the statuses of the period whether open or closed.

The layout:

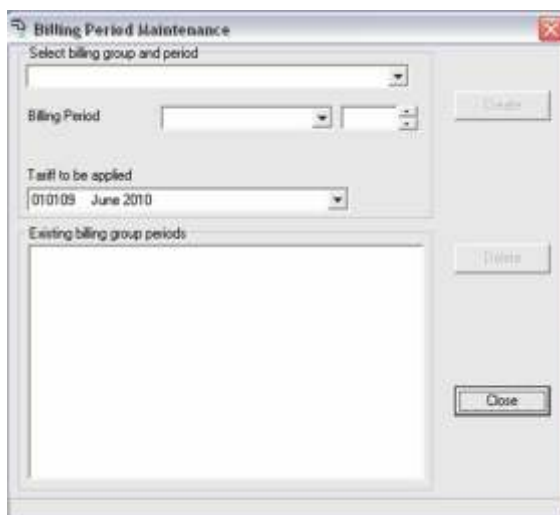


Figure 56: Billing Period Maintenance Screen

- Select the Billing group or zone in which the billing period will be created.
- Select the billing period to be used (usually shows per month).
- Click the create Button to complete the process.
- To delete, Select an existing billing group period and click the delete button to remove the existing billing Period.
- To exit the process, click Close.

4.3. READING BOOK MAINTENANCE

The function allows the user to create a meter-reading book for a billing group in a given period. The reading book has to be prepared and then printed on screen. The same reading book can be printed in hard copy for the meter readers. In reading book maintenance you have the option of printing by route/sub-zone after preparing a book for a zone.

Screen Layout:

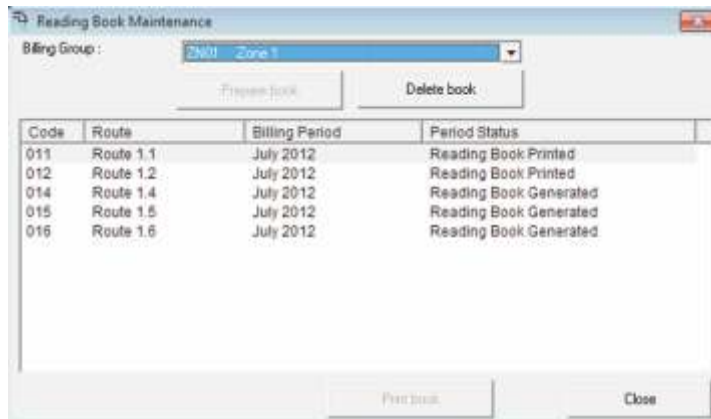


Figure 57: Reading Book Maintenance Screen

- Before creating a reading book, create a billing period in period Maintenance. (Refer to chapter 4.2) The book will appear in the window as seen on the figure 56.
- The next step is preparing the book, select the name of the zone and then click the Prepare Book button to begin the process.
- Click the Print Book button to preview before sending the reading book to the printer.
- Select a reading book and click Delete button to erase the book from the maintenance form.

4.4. EXPORT TO DATA LOGGERS

The function allows the CMS user to send billing groups/zone or routes to a data logger device that will be used in the process of capturing meter readings from the customer's meter location.

A user can prepare the reading book and print it in the system before exporting it to the data logger device.

The user has the option of splitting the reading book if a billing route has a long list of customers which can then be sent to different devices. This can also be used to send lost books to the printer instead of the data logger.

The Layout:



Figure 58 : Export to data loggers Screen

- Select the billing group/zone that is required for export.
- Select the route to send to the data logger.
- Click the send to device button to export the route.
- Click the Split Books button to select the number of records to have in a book that you can send to the data logger or the printer.

4.5. CAPTURE READINGS

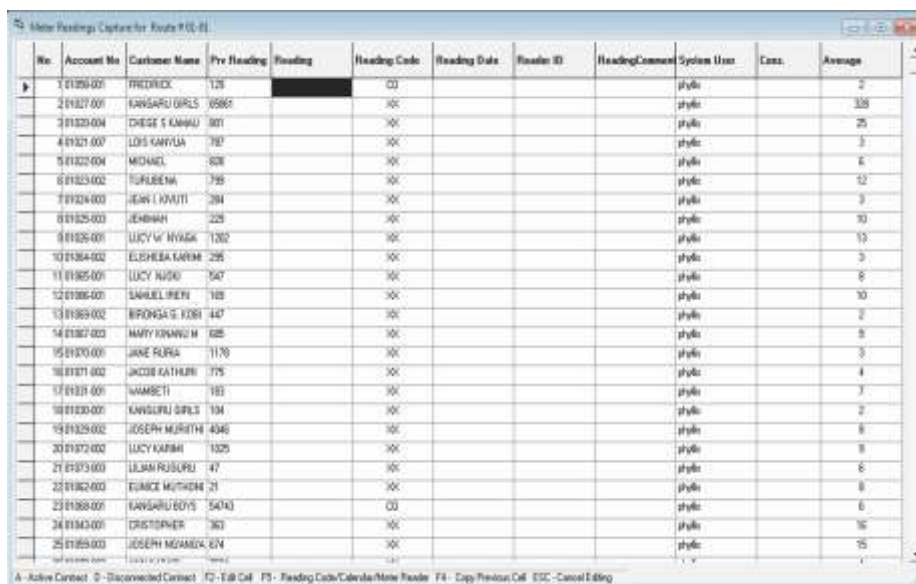
The function allows the CMS user to record readings obtained from the customer's premises by a designated meter reader on a reading date. A worksheet-like screen is presented with the following details customer numbers, names, meter numbers, meter reading column, reading date, reading code, reader ID and comments. In addition there are columns with Previous Readings and Calculated Consumption which help the user to determine negative or very high consumption at an early stage. (Consumption is calculated by entering the current reading less previous reading and made available for display after stopping the capture temporarily.)

The Layout:

Code	Route	Billing Period	Period Status
021	Route # 02-01	March 2014	Reading Book Printed

Figure 59 : Capture Readings Screen

- Click the zone/route to enter readings for, select the designated meter reader and select the reading date.
- Click the Ok button, this will open a worksheet capture screen that resembles a printed meter reading book following the same sequence. Function keys can be used to aid the capture of readings, they can be found at the bottom of the meter reading capture screen.



No.	Account No.	Customer Name	Pre Reading	Reading	Reading Code	Reading Date	Reader ID	Reading Comment	System User	Cons.	Average
1	01050-001	FREDRICK	125		00				philo		2
2	01027-001	KANGARU GIRLS	05887		00				philo		3.58
3	01030-004	CHISE S KANGU	801		00				philo		25
4	01021-001	LOIS KANYUA	787		00				philo		3
5	01022-004	MICHAEL	808		00				philo		6
6	01023-002	TOPIBENA	788		00				philo		12
7	01024-003	JEAN I KIVUTI	284		00				philo		3
8	01025-003	JENBWAH	225		00				philo		10
9	01026-001	LUCY W NYAGA	1202		00				philo		13
10	01084-002	ELDERISA KARIUM	295		00				philo		3
11	01085-001	LUCY NJUGU	567		00				philo		8
12	01086-001	SAHUEL IREH	183		00				philo		10
13	01089-002	WIPONGA S KORI	447		00				philo		2
14	01087-003	MARY KIRANU M	688		00				philo		8
15	01070-001	JAKE KURIA	1170		00				philo		3
16	01071-002	JACOB KATHURU	775		00				philo		4
17	01073-001	WAMBEI	183		00				philo		7
18	01030-001	KANGARU GIRLS	134		00				philo		2
19	01025-002	JOSEPH MURITHI	4346		00				philo		8
20	01072-002	LUCY KARIUM	1325		00				philo		8
21	01073-001	ELIJAH KUBURU	47		00				philo		6
22	01082-003	ELANCE MUTHIGA	21		00				philo		8
23	01088-001	KANGARU BOYS	5403		00				philo		6
24	01042-001	CRISTOPHER	363		00				philo		16
25	01055-003	JOSEPH MENDAKI EN			00				philo		25

Figure 60: Meter Reading Capture Book Entry Screen

From this capture screen, you can enter the readings, reading codes and any additional comments.

The File menu at the top of the window has:

- i) **Print** (Ctrl+P) the Readings that have been already captured/entered can be produced in hard copy.
- ii) **Finish Capture** (Ctrl+S) option to complete the process and move to adjustment capture.
- iii) **Stop Capture Temporally** (Ctrl+X).

You can search and find entries of a customer and their status by using the pipeline number or Serial Number found in the Find Menu.

4.6. IMPORT FROM DATA LOGGER

The function allows the CMS user to retrieve reading records from the data logger device.

The reading records will be inserted into the reading book in the system for further processing.

The steps to follow will include:

- Calculation of consumption
- Consumption adjustments
- and later on Billing Process.

The Layout:

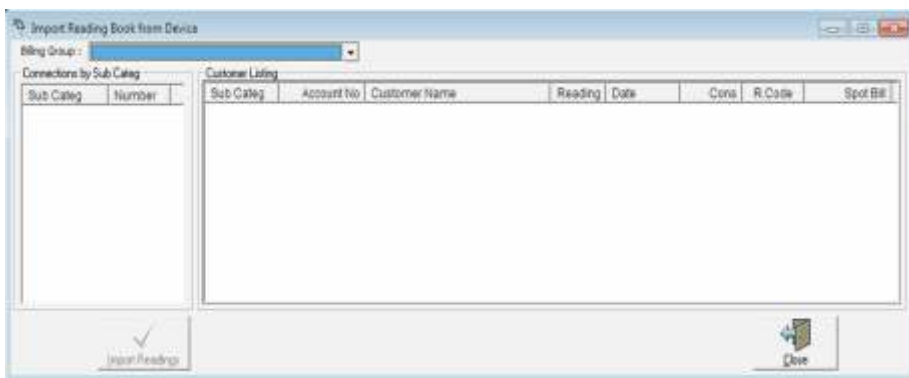


Figure 61 : Capture Readings Screen

- Connect the data logger to the computer
 - Click the drop down menu and select the required billing group zone/route to enter readings for, select the designated meter reader.
 - Click the Ok button, this will open a worksheet capture screen that resembles a printed meter reading book following the same sequence.
- Function keys can be used to aid the capture of readings, they can be found at the bottom of the meter reading capture screen.

4.7. ADJUST CONSUMPTION

The function does a comparison of the utilization of water in a client's site given two meter readings (previous and present readings).

This process can only be initiated if all the customer's readings have been entered and the reading book has been completed. (Refer to chapter 4.5) Anomalies (no readings, negative consumptions, zero consumption, very low or very high consumptions) are noted and the alerts are displayed as seen in figure 63 .

The Layout:

Code	Route	Billing Period	Period Status
503	Route # 50-03	February 2014	Capture Consumption Adjustme
504	Route # 50-04	February 2014	Capture Consumption Adjustme

Figure 62: Process Consumption Control Screen

Select the appropriate billing group/route and click the **Run** button to begin the adjustment capture screen. To continue the process of billing without adjustments click **Skip Adjustments**.

To exit the process, click the **Close** button.

The layout screen:

Figure 63: Consumption adjustment capture screen

The screen above is used to aid the capture of adjustments. It displays a customer's details, previous period details and present period details. In the current period all details can be edited. A new meter reading can be entered to correct erroneous readings capture done on the reading book, the reading code when changed will affect the reading quality. The consumption period is the difference between the previous reading date and the present reading date. The average consumption will contain the consumer's average consumption over the last six months. To use it as the base of billing, copy the content and paste it in the Estimated consumption text box. The Estimated consumption is used to enter the consumption that you would like to bill a customer ignoring the current meter readings or if there were erroneous reading/ no meter readings obtained. Reading comments can be added too.

Click the Next button to adjust the consumption for the next customer. When asked to save the changes, click yes. When you finish the process for all the customers, click **Adjust All** button. The following will appear.

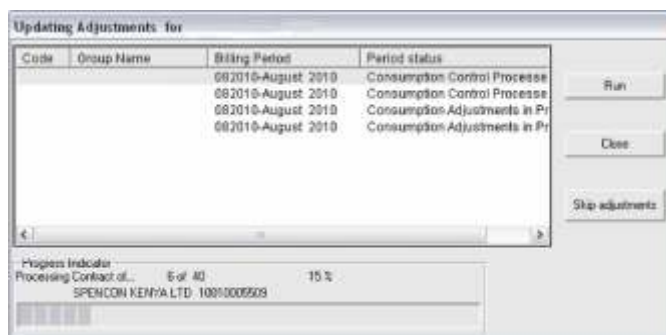


Figure 64: Update Adjustments progress screen

Immediately after the process is complete, a status is shown.



Figure 65: Update Adjustments status

When more errors are found, after the processing, a consumption error report can be produced.

Consumption Errors Report									
BENTON COUNTY Date: August 2020									
Proc. No.	Customer Name	Address	Service No.	Service Date	Service	Qty. Delivered	Units	Cost	Unit Cost
146	164-0000000000000000	164-0000000000000000	730001			699	699	0	\$2.70E+01
700	164-0000000000000000	164-0000000000000000	730008			449	449	0	\$2.70E+01
800	164-0000000000000000	164-0000000000000000	730011			879	886	-7	\$2.70E+01
1000	164-0000000000000000	164-0000000000000000	730015			680	680	0	\$2.70E+01
1400	164-0000000000000000	164-0000000000000000	730015			284	284	0	\$2.70E+01
1700	164-0000000000000000	164-0000000000000000	730019			1699	1699	0	\$2.70E+01
2000	164-0000000000000000	164-0000000000000000	730022			481	481	0	\$2.70E+01
2300	164-0000000000000000	164-0000000000000000	730033			52	52	0	\$2.70E+01
2700	164-0000000000000000	164-0000000000000000	730040			2	2	0	\$2.70E+01
3000	164-0000000000000000	164-0000000000000000	730041			2	2	0	\$2.70E+01
2700	164-0000000000000000	164-0000000000000000	730042			6457	6457	0	\$2.70E+01
3000	164-0000000000000000	164-0000000000000000	730045			0	0	0	\$2.70E+01
3000	164-0000000000000000	164-0000000000000000	730049			545	545	0	\$2.70E+01
1000	164-0000000000000000	164-0000000000000000	730054			1027	1027	0	\$2.70E+01
1400	164-0000000000000000	164-0000000000000000	730058			19	19	0	\$2.70E+01

Figure 66: Consumption Error Report

4.8. CHANGE PREVIOUS READINGS

The function is used administratively to correct any erroneous capture of previous period readings for a customer account.

The Layout:



Figure 67: Modify Previous Reading Screen

- Click the Search button and obtain the customer account in question
- The Customer profile section will be filled with the customer detail
- In the New previous reading section, enter the previous reading and its associated reading date.
- Click the **Update** button to save the changes.
- Click **Cancel** to exit.

4.9. UPDATE READING BOOK

This form is used to compare account numbers found in a billing group with the account numbers in a subsequent reading book under a stated billing period. If the total numbers match then there is no record left out, hence the reading book and the billing group is in order. However, if there is a difference, then the record(s) of the customer accounts are displayed in the customer listing pane.

The layout:

Code	Group Name	Billing Period	Period status
AMAL	AMALEMBA	082010-August 2010	P-Printing of Reading Book Complete
AMSC	AMALEMBA SCHEME	082010-August 2010	P-Printing of Reading Book Complete
KEFI	KEFINCO ZONE	082010-August 2010	Consumption Adjustments in Progress
KEFI	KEFINCO ZONE	092010-Septemb 2010	Q- Reading Book is Generated (Creat...

Account Nos	Reading Book Nos	Difference
87	86	1

Account No	Connection	Customer Name	Pipe Line No	Serial No
☒ 100240	100240	JOHN WAKARU MBURU	100240	5

Only accounts that have [Serial No] can be added to the Reading Book.

Default meter reader: [dropdown] [Update Reading Book] [Close]

Figure 68: Reading Book Maintenance Screen

- Select the Zone and period interested in updating. The Totals of the account numbers and reading book numbers will be displayed and the difference calculated. The difference (customer name(s) and account(s)) will be displayed in the customer listing pane.
- Check (✓) the account(s) to include in the reading book or billing group.
- Select the default meter reader for the selected customer(s).
- Click the **Update Reading Book** button to include the missing account(s).

4.10. UPDATE READINGS

This window is reserved for CMS administrator users who would like to change the readings for the current period. This can only be done after the capture in the reading book has been completed(Refer to chapter 4.4) and Bills must not have been processed.

The layout:

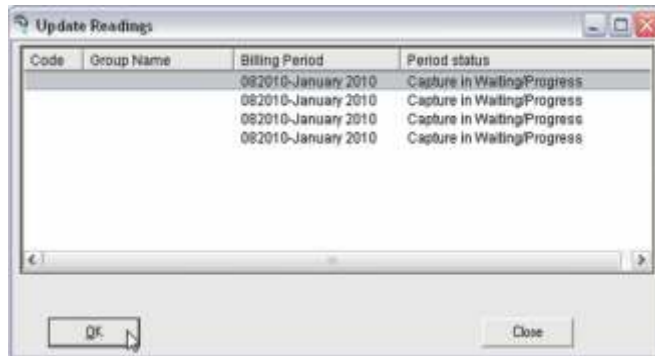


Figure 69: Update Readings loader Screen

- Select the zone and billing period interested in updating.
- Click **OK** button to launch the worksheet for readings capture.

A spreadsheet similar to the capture readings (as seen in Figure 60) will appear and only editing the current readings is allowed. When update is complete, find **Refresh Data** on the **File** menu to save any changes made. Click **Exit** to close the window.

4.11. RE-ORDER ROUTES

This function is useful for ordering routes, walks and any other sequencing processes for the meter reading book. This ordering can be set to fit a meter reader's walks or routes for easy management and efficiency in the meter reading process.

The layout:



Account	Customer Name	Meter No	Route No	Walk No	Address	Account Status	Sub Zone
01940010001	JOEL RUDF	6517078	014	100	11 KERICHO - 2020	Active Connection	Route 1.4
01940020001	BRIDGE MOTORS	0216217	014	200	200 KERICHO - 202	Active Connection	Route 1.4
01940030001	ROBERT S. MUTAI	7041226	014	300	572 KERICHO - 202	Active Connection	Route 1.4
01940040001	MICHAEL MUTAI	4212372	014	400	413 KERICHO - 202	Active Connection	Route 1.4
01940040001	BENJAMIN KEMENI	15021030	014	401	285 KERICHO - 202	Active Connection	Route 1.4
01940050001	DAVID TOWETT	116050267	014	500	217 KERICHO - 202	Active Connection	Route 1.4
01940060001	MICHAEL BWHIE	2002484	014	600	44 KERICHO - 2020	Active Connection	Route 1.4
01940070001	STANLEY KENDUJI	034038	014	700	1900 KERICHO - 20	Active Connection	Route 1.4
01940080001	ESTHER MORAN	05180126	014	800	636 KERICHO - 202	Active Connection	Route 1.4
01940090001	WILFRED ROND SINDIRE		014	900	NONE - 20200	Active Connection	Route 1.4
01940100001	CHARLES NGETICH	00062613	014	1000	893 KERICHO - 202	Active Connection	Route 1.4
01940110001	RUTH KILEL	P185105158	014	1100	152 KERICHO - 202	Active Connection	Route 1.4
01940120001	JOYCE C. TIRIP	7702448	014	1200	1424 KERICHO - 20	Active Connection	Route 1.4
01940130001	AKAMIE OBIAMOLE	67001036	014	1300	201 KERICHO - 202	Active Connection	Route 1.4
01940140001	DAVID W. JACKSON	67001036	014	1400	1702 KERICHO - 20	Active Connection	Route 1.4
01940150001	BENSON KIRAGOMBO	67001036	014	1500	125 KERICHO - 202	Active Connection	Route 1.4
01940160001	PETER M. KIRAGOMBO	67001036	014	1600	300 KERICHO - 202	Active Connection	Route 1.4
01940170001	FERMUS N. JIRIGA	770009	014	1700	790 KERICHO - 201	Active Connection	Route 1.4

Figure 70: Re-order Route Interface

- Select the Zone/Route to re-order, then click the **Load Data** button. There are two options for re-ordering these include:
 - Automatic Re-ordering > Used to re-arrange the sequence according to the walk number or serial number given to the customer in the account details.
 - Manual Re-ordering > A list produced with all the accounts and their order numbers which allows the user to enter the order number sequence.
- Click the close button to save any changes made to Walk Nos.

4.12. METER READERS

This function is used to create a database that consists of employees who are responsible for reading meters from the customer's premises. The meter readers record readings on meter reading sheets printed from the MIS. The function allows one to create modify and delete details of a meter reader.

The layout:



Figure 71: Meter Reader Management Screen

To create a new meter reader:

- Enter all the fields with the necessary information as shown on the layout above. If some fields are missing some information, use the hyphen (-) to signify a N/A field.
- Click the create button to save the details of the meter reader.

To edit details of an existing meter reader:

- Click the search button to select an existing meter reader.
- The information will be displayed on the form, change the necessary details then click the Modify button.

To delete a meter reader:

- Click the Search button to select the already existing meter reader.
- The information will be displayed on the form.
- Click the delete button to erase the meter reader from the database.

4.13. LINK MASTER METERS

This function is used to assign master meters that are used to feed bulk water to consumers in an area or zone. In the window you can assign master meters to other master meters from a high level to a deeper level in the form of a hierarchy. Each low level meter can serve a number of customer connections.

The layout:

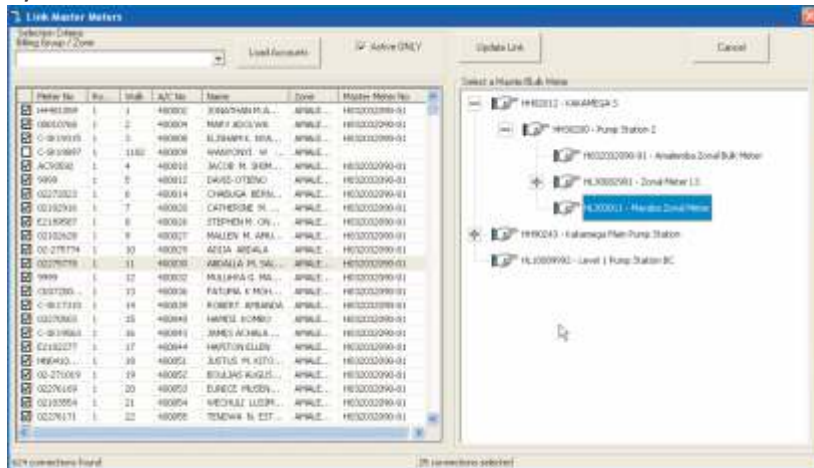


Figure 72: Link Master Meters Screen

Click and select the billing group/zone/route at the selection criteria section and click Load Accounts button. To load all accounts, uncheck the Active only check box.

On the Right side of the form, a list of all created master meters is shown. This list is created in master meter maintenance when specifying the levels and distributor meter (feed meters). Expand the options till the desired master meter level is achieved.

Select the Master meter name and on the left side of the form, check (✓) the connections served by the selected master meter.

Right click on the customer list view to get a menu that can be used to select all records, un-select all, inverse selection, remove checked and un-checked records. Click Update Link button to complete the process.

4.14. CAPTURE MASTER READINGS

This function is used to record the readings for any master meter installed in an area/zone/route. The master meters can be read at an hourly/daily rate based on the company policy. This module will facilitate capturing the readings into the system for purposes of analysing the UfW.

The layout:

Reading Date	Reading	Meter Number	Reading Comments	Last Updated on	Updated By
7/1/2010	30000		ok	10/7/2010 9:45:34 PM	SA
7/2/2010	32510		ok	10/7/2010 9:46:40 PM	SA
7/3/2010	33340		ok	10/7/2010 9:46:53 PM	SA
7/4/2010	34500		ok	10/7/2010 9:47:02 PM	SA
7/5/2010	35790		ok	10/7/2010 9:47:07 PM	SA
7/6/2010	36710		ok	10/7/2010 9:47:09 PM	SA
7/7/2010	37010		ok	10/7/2010 9:47:15 PM	SA
7/8/2010	38230		ok	10/7/2010 9:47:20 PM	SA
7/9/2010	39130		ok	10/7/2010 9:47:39 PM	SA
7/10/2010	30290		ok	10/7/2010 9:47:43 PM	SA
7/11/2010	31720		ok	10/7/2010 9:47:48 PM	SA
7/12/2010	32610		ok	10/7/2010 9:47:52 PM	SA
7/13/2010	33000		ok	10/7/2010 9:47:56 PM	SA
7/14/2010	34100		ok	10/7/2010 9:48:22 PM	SA
7/15/2010	34990		ok	10/7/2010 9:48:26 PM	SA
7/16/2010	35010		ok	10/7/2010 9:48:31 PM	SA
7/17/2010	36710		ok	10/7/2010 9:48:40 PM	SA
7/18/2010	37320		ok	10/7/2010 9:48:46 PM	SA
7/19/2010	38100		ok	10/7/2010 9:48:54 PM	SA
7/20/2010	38790		ok	10/7/2010 9:49:37 PM	SA

Figure 73: Capture Master Meter Readings Screen

Search or enter the master meter number which will display the location. Select the billing month that the readings are going to be captured.

The reading sheet resembles the worksheet found in capture readings. You can stop the capture temporarily and resume at a later time. When reading sheet is complete, select finish capture to end the process.

4.15. PROCESS UfW

UfW stands for Un-accounted for Water. This is calculated by obtaining the difference between the consumption of the master meter and customer connections linked to the master meter. It enables the technical department in the company detect burst pipes and leakages among others.

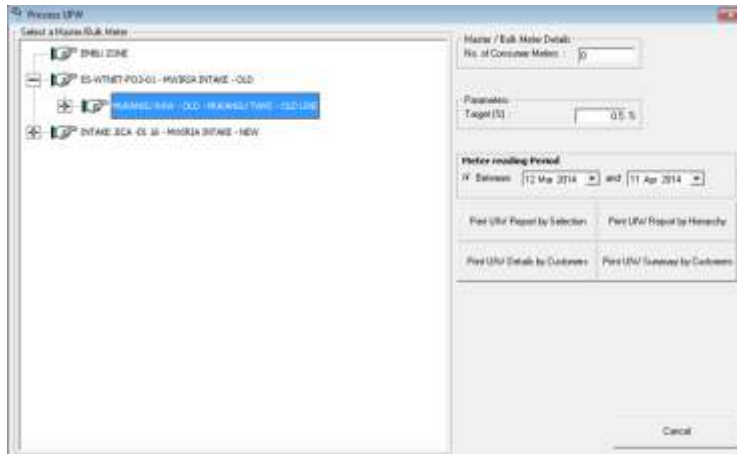


Figure 74: Process UfW Screen

To view the report, click on process UfW on the metering menu and select the appropriate options to suite the report you would like to produce.

4.16. DISCONNECTION ALERT

This is a report builder that generates information about suspected illegal connections under a billing period for customers in specified zones. The report contains details such as the serial number, account number, connection number, customer name, meter number, periodic readings (PR), consumption, disconnection date and disconnection reading (DR).
Difference =PR-DR

The layout:



Figure 75: Disconnection Alert Report Builder

- Select the zone and the billing period that you want to view the disconnection alerts.
- Click the print icon button to preview the report before sending it to the printer.
- The report will display customers' connections that are suspected to be illegal hence require further investigations.

4.17. GROUP PERIOD FOLLOW UP

The function shows the user summary information of the progress of the Billing period, status, and the dates.

The summary of this information can be viewed either by group or period. This is a function to enable a quick view of readings of customers per zone.

4.17.1. *Summary by Group*

This function displays progress of a billing period for a Billing group/ Zone.

The layout:

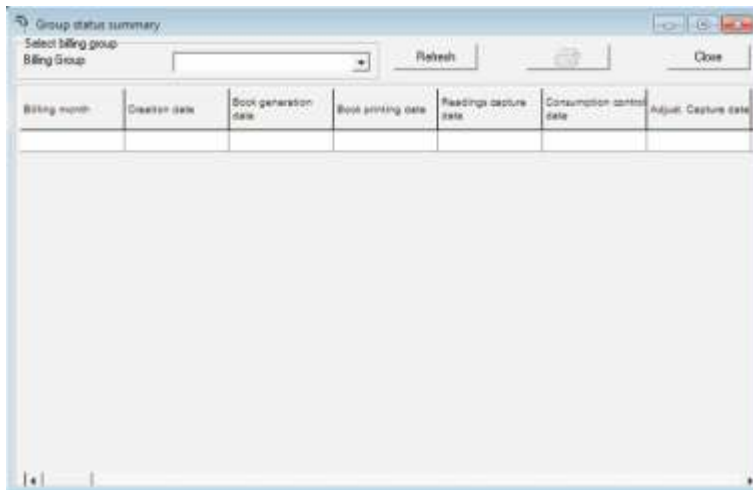


Figure 76: Group Status Summary Ordered By Billing Group Screen

- Select the Zone which will be used to obtain the billing period statuses.
- Click the **Refresh** Button to view the list.
- Click the print icon button to preview a copy before printing to the printer.
- To view the progress of the billing period of other billing groups select another Billing Group then click the refresh button.
- To exit, click the **close** button.

4.17.2. *Summary by Period*

This function displays the progress of all the zones for a specified billing period (Month and year).

The Layout:



Figure 77: Status Summary Ordered By Billing Period Screen

- Select the month and year you would like to view the progress.
- Click the **Refresh** button to display the records
- Click the **print** button to preview the results or send to a printer.

4.17.3. *Display Readings*

This function shows the billing group/ zone names which can be selected to open a window that displays meter readings for each customer in the zone.

The layout:

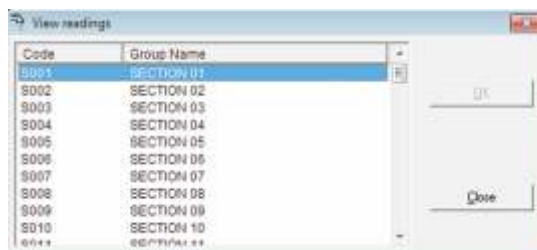


Figure 78: Display Readings Selector Screen

- Select the group name and click the ok button to proceed.
- A new window with a customer name and account details will open with meter readings since the start of meter reading for the connection. Click Next button to view other accounts.

4.18. REPORTS

The reports function creates a summary of information that can be printed out for analysis purposes. The reports encompass operations that have been carried out in the metering menu. The reports found here include:

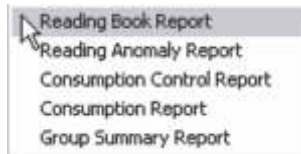


Figure 79: Metering Reports Menu

4.18.1. *Reading Book Report*

This report builder is used to print the readings captured for a billing group and period.

The layout:

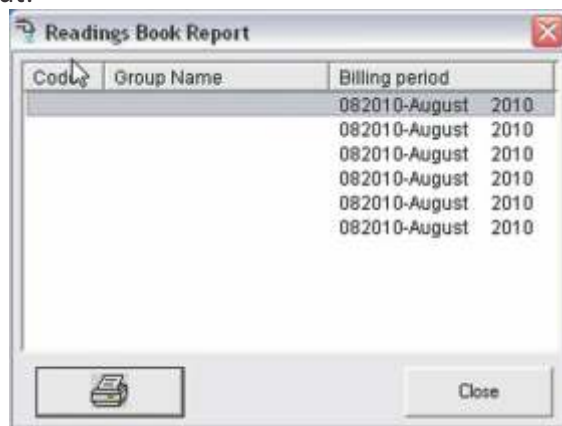


Figure 80: Reading Book Report Builder

Select the billing period and click the Print icon button to view the meter reading entries and print them out on hard copy paper.

4.18.2. *Reading Anomaly Report*

This report produces a summary of irregularities found in the metering process for a billing group/ zone in a billing period. The report's success is highly dependent on the reading codes captured by meter readers while recording meter readings from customer's premises.

Layout:



Figure 81: Report Reading Anomalies Report Builder

- Click the billing period and group name to view a summary of the anomalies.
- At the right side of the form select the criteria to display in the report (Different Serial number or No meter present or no reading)
- Click the print icon button to preview the report or send the report to a printer.

4.18.3. Consumption Control Report

This function produces two reports: Consumption Error report and Consumption Adjustment report.

The layout:

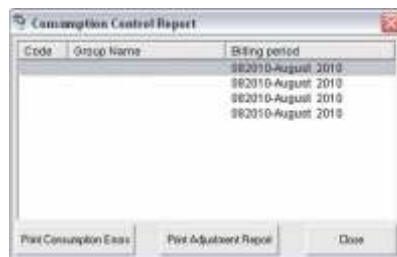


Figure 82: Consumption Control Report Builder

- Select the billing group name and the billing period.
- Click the Print Consumption Errors to preview erroneous readings entered in the reading book. These errors can be found if adjustments were skipped or some accounts were not given the correct reading hence having a negative or zero consumption.
- Click the Print Adjustment Report to view modifications made on the meter readings in the Adjust consumption module.

4.18.4 Consumption Report

The following report includes the display of consumption that have been worked out using the previous and present readings for a zone.



Figure 83: Consumption Report Builder

- Click the print icon button to preview the report or send the report to a printer. The report has the account name, number, reading date, meter reading calculated consumption and average consumption. It also give a total of all consumption for a period.

4.18.5. Group Summary Report

The report is handy when it comes to showing the group status as at a billing month.

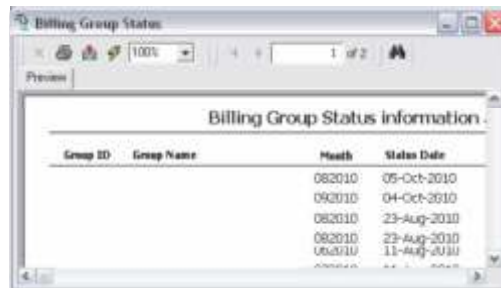


Figure 84: Consumption Control Report Builder

4.18.6. Meter Reading Routes Report

The report provides a comprehensive list of customers grouped in a meter reading route. It also displays the customer account status.

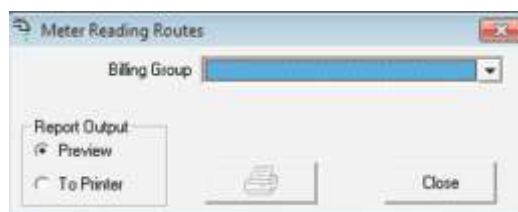


Figure 85: Meter Reading Routes Report Builder

5. BILLING

5.1. BILLING GENERAL

The menu contains functions that deal with statement processing for customers. It provides you with the option of bill simulation, processing bills, creation, printing, displaying, bill adjustments, bill cancelations, bill distribution, reports and processing draft bills. All these are shown bellow as they appear on the menu:

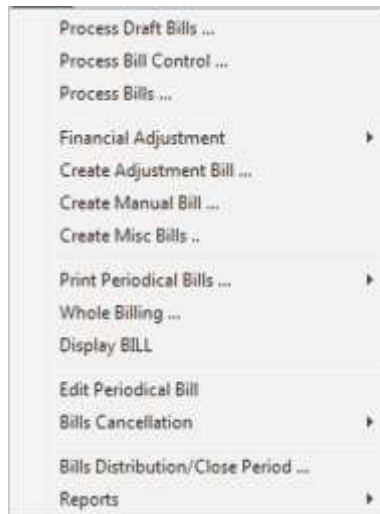


Figure 86: Billing Menu Content

5.2. PROCESS DRAFT BILLS & CONTROL

On selecting Process Draft Bill for a zone, the system processes pro-forma bills that do not update the customer's account, but simulate actual amounts to be billed for consumers according to services applied for. In Bill Control, the billing group is checked against the tariff the consumer is placed to determine if the amount to be billed exceeds the stipulated range for that tariff. When billing anomalies are found, ("zero" bills, -ve bills, out of range bills for the particular consumer category, very large bills, and unavailable tariff lines) the system compiles them into a billing anomaly report which can be displayed and printed. Draft Bills can be done for a zone that has undergone consumption control and has had billing adjustments captured.

Layout:



Figure 87: Process Draft Bills Screen

- Select the zone and click the process button.
- When complete, select the order by option and click the print button to display the report with each customer and the amount to be billed.

In bill control, a zone is processed to check for high consumption errors. When errors are generated an error report with all anomalies is generated. When no errors are found the status of the billing group is changed. This allows you to print bills from the system (Refer to Chapter 5.3)

The Layout:

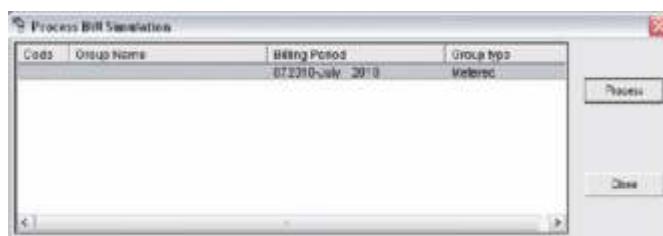


Figure 88: Process Bill Control Screen

5.3. PROCESS BILLS

The function allows a user to launch bill processing for every customer in a billing group after undergoing consumption control. The system takes the consumption for the customer and using the applied tariff, calculates the bill for the customer. The bills are created and stored in the system. The user selects a billing group that has gone through draft bills and bill control. This is followed by the distribution of the bills to the clients.

The layout:

Code	Group Name	Billing Period	Group type
		07 2010-July 2010	Metered
		07 2010-July 2010	Metered

Billing Date: 13 Oct 2010 Distribution Date: 14 Oct 2010

Figure 89: Process Bills Screen

- Click the billing group/ zone and period expected to be billed.
- Select the billing date and the date at which the bills are expected.
- Click the process button to begin the procedure to be distributed.

5.4. FINANCIAL ADJUSTMENT

This function gives a user the ability to create a bill for a customer through raising a Debit Note or Credit Note which is certified through an authorization process. The process involves the relevant authority logging into the system with their credentials and permitting the financial adjustment to be effected on the customer's account after approval.

A debit note will increase (+) while a credit note will decrease (-) the amount on a customer's statement.

Figure 90: Prepare Adjustment Screen

- Enter the customer's account number and click the ok button.
- Select the type of adjustment (Debit / Credit note) and enter the description, quantity, unit price and account affected. Go ahead and add a short description and a long description about the bill.
- Click the Create Adjustment button to record this in the system.

In adjustment authorization screen, the user with relevant authority must log in, select and approve the adjustment.

Figure 91: Adjustment Authorization Screen

- Click show all pending adjustments button to display a list with all the available adjustments.
- Click on the relevant adjustment in question and click recommend or update account button to update the customer account.

In account adjustment reports, a user can specify various options to get varied results in the analysis reports.

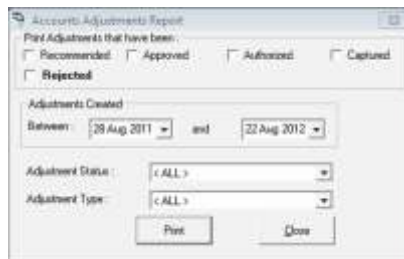
The image shows a software window titled "Accounts Adjustment Report". Inside, there's a section "Print Adjustments that have been:" with four checkboxes: "Recommended", "Approved", "Authorized", and "Captured". Below this is a "Rejected" checkbox. The "Adjustments Created" section has a "Between:" label followed by two date pickers showing "29 Aug 2011" and "22 Aug 2012", separated by an "and" label. Below the dates are two dropdown menus: "Adjustment Status" and "Adjustment Type", both currently showing "< ALL >". At the bottom are "Print" and "Close" buttons.

Figure 92: Accounts Adjustment Report Builder

- Select the appropriate options and click print button to show the results, otherwise click close.

5.5. ADJUSTMENT BILL

This function gives a user the ability to create a bill for a customer by providing the correct readings of the water meter. By entering the reading the system automatically calculates the bill to determine whether to charge the account with a credit or debit. The user can bill using calculated consumption (by providing the current reading) or estimated consumption.

The screenshot shows a software window titled "Create Adjustment Bill". It features a search bar for "Account No." with a magnifying glass icon and an "OK" button. Below this, there are radio buttons to select the "Adjustment Type" as either "DEBIT NOTE" (which is selected) or "CREDIT NOTE". A section labeled "Bill Details" contains fields for "Account ID", "Account No.", "Name", "Metering Indicator", "Previous Reading", "Previous Reading Date", "Previous Cons.", "Previous Base", "Account Balance", and a "Status" field. The "NEW BILL" section includes a "Billing period" dropdown, a "Meter Reading" input field, a "Reading Date" dropdown (currently showing "23 Aug 2012"), a "Reading Code" dropdown, a "Meter Reader" dropdown, and a "Reading Comments" text area. The "Billing Method" section has two radio buttons: "Calculated Consumption" and "Estimated Consumption", along with a checkbox labeled "Save Readings as Previous Reading for next Billing". At the bottom, there is a "Billing Comments" text area and two buttons: "Save Bill" and "Close".

Figure 93: Create Adjustment Bill

- Enter/Search the customer's account number and click the ok.
- Enter the details of the new bill. e.g Billing period, Meter reading, reading code, Meter Reader Name and any reading comment.
- In the Billing Method the user has the option of using Calculated consumption or estimated consumption for billing.
- Enter the billing comments and Click the Save Bill button to effect the customer account with the adjustment.

5.6. MANUAL BILL

This function allows the user to create a bill (debit) that will be charged on a users account. The system will calculate the bill using the tariff structure. This function can be used in cases where the customer would like to clear an account before they move out. By providing the final meter readings (actual/ estimate) the system shall calculate a bill or charge a customer for illegal water consumption.

Figure 94: Create Manual Bill Screen

- Click the search icon button to obtain the Account No
- The Bill details will be displayed in the form. Note this section will be for display purposes only.
- In the New Bill section, select the billing period, enter a new meter reading, the reading date, the reading code and reading comments. Select the Billing method to be used (Calculated / Estimated). You can choose to save the readings for next billing period if you provided a true meter reading for the account.
- Click the Save Bill button to store the new bill for the customer.

5.7. MISCELLANEOUS BILL

This function allows the user to capture and bill extra services provided by the company to customers and non-customers. E.g. Exhausting, Dumping and Tanker services. This results in an invoice that the customer can pay against.

Figure 95: Create Miscellaneous Bill Screen

- Click the drop down and select the service to bill
- For a new customer type in the name, for a customer that has been billed before click the drop down menu and select their name.
- Next Add the address line, reference number and select the invoice date.
- In the line type select the appropriate service name from the list. Enter the Unit price and quantity for the service line selected. Next, type the customized description and click **Add Line**. Add as many lines as required for the service.
- Click the **Update** button to create and preview an invoice before sending to a printer.
- Reprinting invoice button will require entering the invoice number to facilitate previewing an already created invoice in the system before sending to a printer.

5.8. PRINT PERIODICAL BILLS

This function facilitates the production of created bills in softcopy or hard copy in preparation for the distribution process. The user has varied options of printing the bills. They could select the periodical bills, the selected/copy bills, display bill option or print bill from display account details in customer.

5.8.1. *Printing Periodical Bills*

This function enables the user to print out bills for different billing groups ordered by the date that the billing process was carried out. management menu.

The Layout:

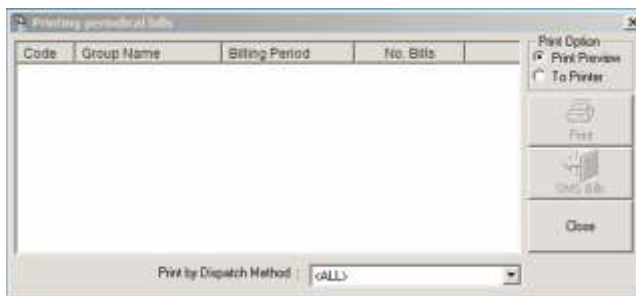


Figure 96: Printing Periodical Bill Screen

- Select the Billing Group and its subsequent billing date.
- Select the print option (Print Preview displays a copy for the user while To Printer option sends the bill to the printer directly)
- The SMS bills Button can be used to send bills for a group/zone /route by sms once those consumers have valid mobile numbers and their bill dispatch method is set to SMS.
- Click the Print icon Button to preview or print the bills.

5.8.2. *Selected Bill*

This function is used to print bills already prepared from a billing group with the option of printing a bill for one, all or some consumers. The account number, connection number and customer name information is made available on the right hand side of the form for the selection process.

The layout:

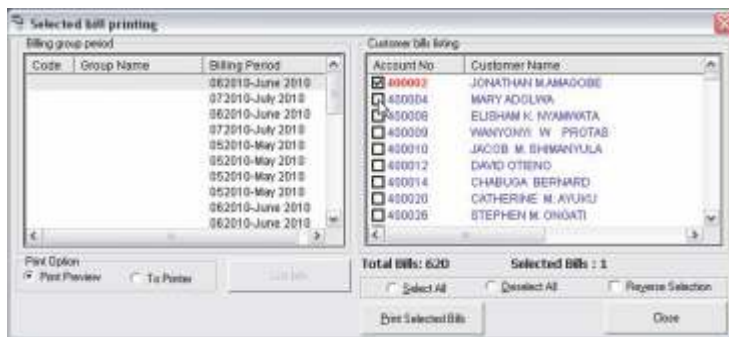


Figure 97: Print selected Bills Screen

- Select the billing group/zone that has its bills ready.
- Click the List bills button to view all the bills for the customers on the right hand side of the form.
- You can select a customer account number or name by checking (✓) the box next to the account number. Use options such as (Select All or Deselect All) to ease selection. The reverse check box can be used to uncheck the checked accounts and check the unchecked accounts and vice versa.
- Click the Print Selected Bills button to print only the checked bill(s). Click close to terminate or end the process.

5.9. E-MAIL PERIODICAL BILLS

This function allows the user to send bills created under the billing period to the customers who have valid email addresses captured into the system. For customers to appear in the e-mail group, they need to have provided the correct e-mail address and it must be captured in the system. If the bill Dispatch method for a customer is set to e-mail, the system shall include the customer's bill in a group that will appear in the screen below set for delivery electronically.

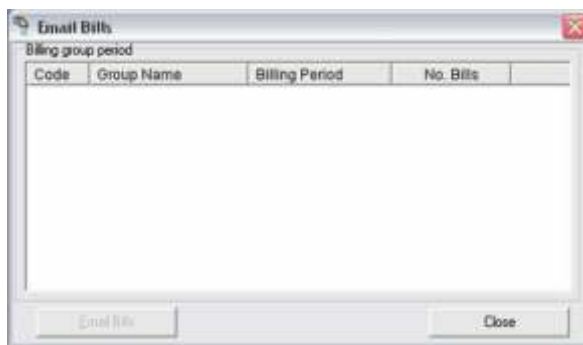


Figure 98: Send Bills via E-mail Screen

- Select the required group name/zone/route displayed on the screen above with customers whom you would like to send bills via e-mail to.
- Click the E-mail bills button to queue them up in an email server for delivery.
- Click Close to exit the screen

5.10. DISPLAY BILL

The function is used to show details of a bill for verification purposes or viewing purposes. The user selects the display bill function and is taken to a dialogue that facilitates the selection of a bill to be displayed. There is an additional option of printing from this form.

The layout:

Figure 99: Display Bill Screen

- Select the search icon button and obtain the required bill.
- The customer profile section will display the information about the owner of the bill.
- Click the Show Details button to expand the view. The following is what will be shown.

Description	Units	Unit Price	Total	Account
METER RENT	1.00	50.00	50.00	METER RENT
6 m3 @ 45.00	6.00	45.00	270.00	WATER CHARGE
13 m3 @ 67.50	13.00	87.50	877.50	WATER CHARGE
SEWER CHARGE (75%)	1,147.50	0.75	860.63	Sewerage

Buttons at the bottom: Print Bill, Email Bill, Close, Hide Details <<

Figure 100: Detailed Bill information Screen

- To print the statement show in the bill line section select between the printer types then click the print statement button. If the client has a valid e-mail address, the bill can be forwarded to the address by clicking the e-mail bill.

5.11. EDIT PERIODICAL BILL

This function gives a user the ability to change a bill for a customer by providing the correct readings of the water meter for one billing period bill. By entering the reading the system automatically calculates the bill and replaces the old bill with the recalculated new bill. This can be done only if the periodical bills have not been distributed from the system. The user can bill using calculated consumption (by providing the current reading) or estimated consumption.

Figure 101: Create Adjustment Bill

- Enter/Search the customer's Bill ID or use the search button to obtain it and click the ok.
- Enter the details of the new bill e.g Meter reading, Reading date, Reading code and Billing Method.
- In the Billing Method the user has the option of using Calculated consumption or estimated consumption for billing.
- Enter the billing comments and Click the Save Bill button to effect the customer account with the adjustment.

5.12. BILL CANCELLATION

The function facilitates the cancellation of a bill due to an erroneous entry. The user can annul one bill at a time or cancel them in bulk through the various options explained below. When a bill is cancelled it is no longer stored in the system rather it will be stored as an event in the audit trail.

5.12.1. *Cancel Bills*

This option is used to annul one bill at a time. The layout is as follows:

Figure 102: Cancel Bills Screen

- Select the search icon button to obtain the Bill ID. (Refer to Search Criteria)
- The customer profile will be displayed for the user to ascertain that it is indeed the owner of the bill.
- Click on the Cancel Bill Button to completely erase the customer's bill.

An expanded view of the customer bill is displayed as shown below:

Description	Status	Unit Price	Total	Account Name
Recommendation fee		1.00	500.00	500.00 QUANTIS

Figure 103: Extended information about a Customer's Bill

- This section displays information about the bill, where comments can be added and the bill cancellation date selected.
- Click the Cancel Bill button to cancel the bill.

5.12.2. *Batch Bill Cancellations*

The function allows the user to cancel all the bills of a billing group for a billing Period. This only happens in very exceptional circumstances. Since this is a crucial function, it will only be available for system administrators and those that the company deems fit to have. It could happen if all the readings were entered or captured wrongly or the processing was done using a wrong tariff.

The layout:

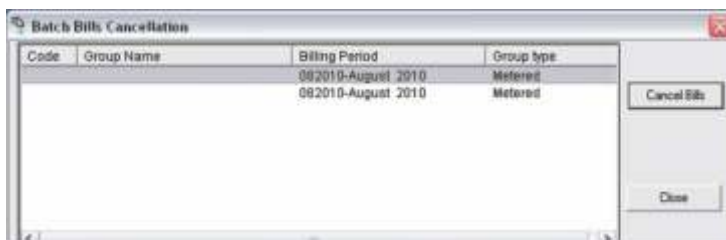


Figure 104: Group Bill Cancellation Screen

- Click the billing group name/zone/route and the billing period you want to terminate.
- Click the Cancel Bills Button to annul the whole billing period.

5.13. BILL DISTRIBUTION

This function is used to record the date that the billing period will be closed. It has the same meaning as closing a billing period in the system. It can be done when a new billing period is about to be created for a zone/route/area.

The layout:

Code	Group Name	Billing Period
		082010
		082010

Code	Group Name	Billing Period
		072010

Bills distribution date: 18 Sep 2010

Update Close

Figure 105: Bill Distribution and Previous Period Closing Screen

- Select the billing group period provided in the left hand side of the form. The billing periods available here are ready for distribution.
- Select by checking (✓) the bills distribution date.
- On the left hand side of the form click the Billing Group Name/Billing Period presented to distribute.
- Click the Update button to distribute the period.

5.14. REPORTS

The report builder found under billing sub-menu is used to provide a summary of the billing process conducted. The summary contains all the billing groups and the categories of customers. It also records how much each service generates, in terms of, money remittances to the company. The report builder provides a detailed report on all the bills created in a selected billing period. This displays information such as the billing group, period, billing date and the total bill. It provides details of all the customer bills.

5.14.1. *Billing Detailed & Summary*

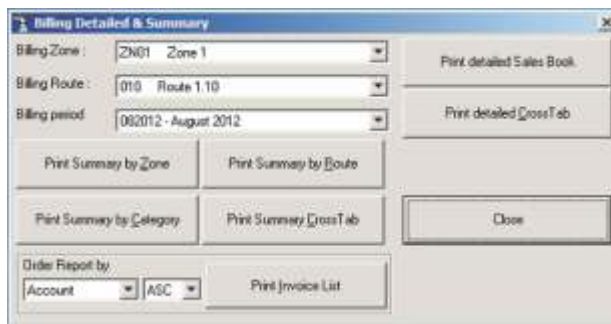


Figure 106: Billing Detailed and Summary Report Builder

- Select the billing group/Zone/Route name you want to generate a summary by period report.
- Select the Billing Period available for the selected billing group.
- You can order the Invoice List report by various criteria (Account, Bill Amount, Consumption or Customer name) from Highest to Lowest (Descending) or Lowest to highest (ascending) order. Click the Print Invoice list button to preview a list of all customers invoiced per the selection criteria used.

- Clicking the various print summary buttons available on screen will produce reports varied reports. One can print summary reports by billing group, zone, route and category.
- When you click on the Print detailed sales book, a report containing amounts collected for each billing group/zone/route broken down as water, rent & sewer and any other charges is displayed.
- When you click on the Detailed Cross-tab, a report containing bills, payments/adjustments, charges and balances carried forward are displayed per zone.

5.14.2. *Print Miscellaneous Invoices*

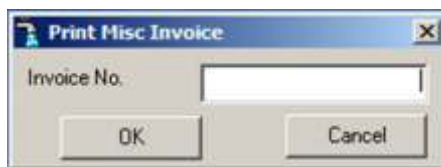


Figure 107: Print Miscellaneous Invoice Screen

- This feature is used to retrieve and re-print invoices generated from miscellaneous bills. (Refer to Chapter 5.7)
- Enter the Invoice number and click Ok to preview the invoice before queuing it on the printer.

5.14.3. *Miscellaneous Invoices Reports*

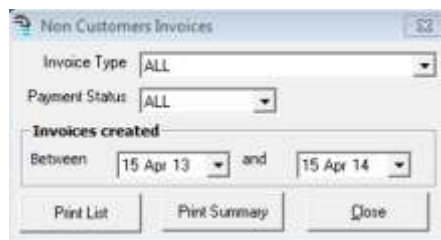


Figure 108: Miscellaneous Invoice Report Screen

- This feature is used to analyze the non-customer invoices generated from miscellaneous bills between a period.
- Select the Payment Status and the valid period for the report.
- Click the Print List button or Print Summary to view various reports.

6. | Revenue

6.1. REVENUE GENERAL

This function has sub-menu items that touch on payments used by a cashier. These payments come from clients who are responding to bills received or other services rendered. This function contains several facilities which include periodical payments, connection payment, deposits, service payments and many more. Furthermore, there are modules for refunding and cancelling payments when it is absolutely necessary. The various components are further explained below.

The following is the layout of the menu:

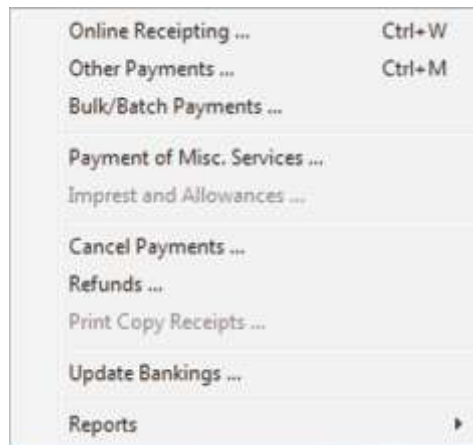


Figure 109: Revenue Menu Items

6.2. ONLINE PAYMENT

This function provides the cashier with a form that record payments for a selected service offered by the company. It has been assigned with shortcut key for quick accessibility since it is a function that is specialized for the cashier. The shortcut key is Ctrl+W. It is mostly used in real time, when a customer walks in to the cashier desk to make a payment.

The layout:



Figure 110: Online Payment Screen

- Click the search button or type in the customers account number. using the customer search engine. Loading the customer's account gives the previous and current water bill and the total account balance. The payment date & Receipt number will be filled in by the system.
- Enter, the amount and select a mode of payment. Include a reference no if the mode of payment is by cheque, bank slip or swift transfer. Click the Add Button to insert the amount as a list.
- Enter the Amount to be received into the system since the customer could give an amount that would require change to be given. The change due shall then be displayed on screen for the cashier.
- When the transaction is done, click the Update button to effect the amount received to credit the customer's account.

6.3. OTHER PAYMENTS

This function is used to capture payments for other services other than payment of periodic bills. These services include Deposits, dis/re-connection fee, new meter, contract fee etc. The shortcut key for this function is Ctrl+M.

The layout:

Figure 111: Other Payments Screen

- Select the Capture auxiliary/ other payments section drop down menu to select the correct service a customer is paying for.
- Select the customer account number by clicking the search icon button or enter it. This will display the customer profile.
- On the same form a new section will appear which will contain details about the amount to be paid for the selected services according to the tariff structure. A feature for topping up a deposit and printing receipts online is available for use.

Figure 112: Additional Information about Payment to be Made

- Click the update button to reflect the payments received on the customer's account.

6.4. BULK / BATCH PAYMENT

This function, as the name suggests, is used to capture payments in back-office. This can happen when the MIS is not working due to power black-outs, server crashes etc. All recorded receipts can then be entered at once

The layout:

Figure 113: Bulk Payment Screen

- Click the search icon to obtain the customer account number. Enter the payment date and receipt no, amount and mode of payment.
- Click Add button. Repeat the process until all the payments captured by hand are captured in the system.

NOTE: that a list of all entered payments will only be displayed on customer's accounts after the UPDATE button is clicked.

6.5. PAYMENT OF MISC. SERVICES

The function facilitates the receiving of payment by non-customers for services provided by the company . It requires that a miscellaneous bill be created in the system before availing the invoice number to the cashier. (Refer to Chapter 5.7)

The layout:

The screenshot shows a software window titled "Misc Receipting". It contains several input fields and buttons:

- Top row: "Invoice No." (text box), "Amount" (text box), "Mode of Payment" (dropdown menu showing "CASH"), and "Ref. No." (text box) with an "Add" button.
- Second row: "Invoice Ref. No.", "Customer Name", "Description", "Invoice Date", and "Invoice Amount".
- Third row: "Invoice No.", "Ref. No.", "Customer Name", "Date", "Inv Type", "Mode", and "Amount".
- Bottom section: "Enter", "Clear", "Exit" buttons; "No. of Lines" (text box) and "Batch Total:" (text box) with a currency symbol; "Amount Received" (text box) and "Change Due:" (text box) with a currency symbol; "Update", "Reprint Receipt", and "Close" buttons.

Figure 114: Miscellaneous Payment Screen

- Enter the Invoice no of the non customer in the space provided
- Use tab key on the keyboard to move the cursor to the next text-box.
- Enter the amount of the bill and select the mode of payment.
- Enter the Reference No if required and click the Add button
- Enter the Amount received from the customer.
- Click the update button to effect the payment for that account.

6.6. CANCEL PAYMENT

The function facilitates the annulment of a recently captured payment. It can be annulled when a payment has been made wrongly or if the payment was not received and a reflection was made on the system that a payment was received. It can also happen if a cheque given bounced, fraudulent posting was carried out or posting was done wrongly.

The layout:

Figure 115: Cancel Payments Screen

- Click the search icon button to obtain the payment number. This will load the customer profile details which will be displayed for confirmation that the payment number belong to that customer.
- The details of the payment to be cancelled will be displayed. Select the reason for annulling the payment and the date that the payment is being cancelled.
- Click the Cancel Payment button to complete the annulment.
- Click the Close button to exit the function.

6.7. REFUNDS

This is an important function that facilitates compensation when a customer decides to terminate their contract with the company.

The deposit returned is determined by the tariff category of the consumer and if indeed they had paid a deposit at the initiation of the contract with the company. A partial refund is possible and can be used when a customer has some unpaid bills which in total are less than the deposit amount. The balance is then refunded to the customer.

The layout:

Figure 116: Refund Payments Screen

- Select the type of refund to be made.
- Click the search icon button to obtain the customer account number to refund. Check (✓) the option to indicate if it is a partial refund.
- The customer profile details will be displayed. Additional details about payments will be displayed as shown below:

Figure 117: Payment Details Screen

- This will show if the account has refundable amounts. At the Payment Details section enter the Reference number, select the cash account and the billing period. The system will display the refundable amount.
- A partial refund will automatically deduct the owed amount from the total amount to be refunded.
- Click update button to reflect the changes on the customer's account.

6.8. UPDATE BANKINGS

This is a function that is used to consolidate the daily amounts collected by the cashier(s) that they submit for banking purposes.

It can be used to generate reports and help in showing the various modes of payments received in the company.

A report can then be generated from the data collected in this form.

The layout:

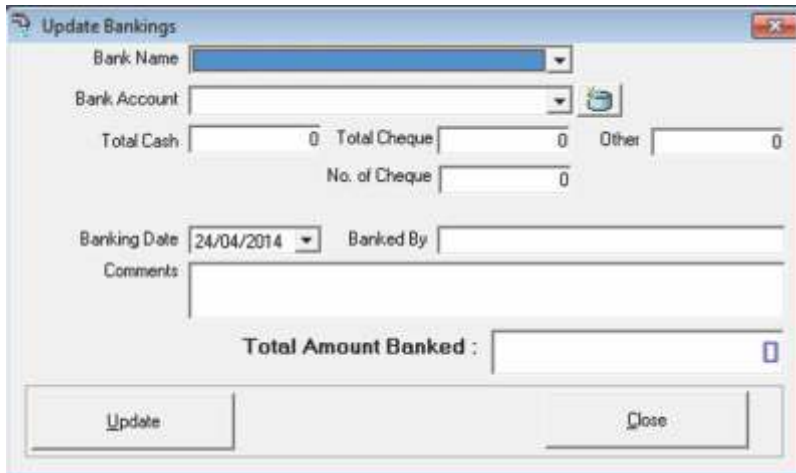
The image shows a software window titled "Update Bankings". Inside the window, there are several input fields: "Bank Name" is a dropdown menu; "Bank Account" is a dropdown menu with a small cylinder icon to its right; "Total Cash" is a text box containing "0"; "Total Cheque" is a text box containing "0"; "Other" is a text box containing "0"; "No. of Cheque" is a text box containing "0"; "Banking Date" is a dropdown menu showing "24/04/2014"; "Banked By" is a text box; "Comments" is a large text area; and "Total Amount Banked :" is a text box containing "0". At the bottom of the window, there are two buttons: "Update" and "Close".

Figure 118: Update Bankings Capture Screen

- Select the Bank Name from the drop down menu.
- Select the subsequent Bank Account from the drop down menu or click the cylinder Icon on the right to add a new bank account.
- Enter the total value of the cash, cheque, number of cheques and any other form of payments received.
- Enter the banking date and the name of the person responsible for banking the amount. Any other comments can be entered in the comments section.
- If the total amount entered tallies with the figure in the screen, go ahead and click the update button to store this in the system.

6.9. REPORTS

The function is used to show the different types of payments made to the company and to provide data for analysis about the payment patterns for a billing group. The following is a sample of reports available:

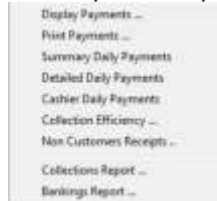


Figure 119: Payment Reports Menu Items

6.9.1. *Display Payments*

This report is used to show a customer payment history.

The layout:

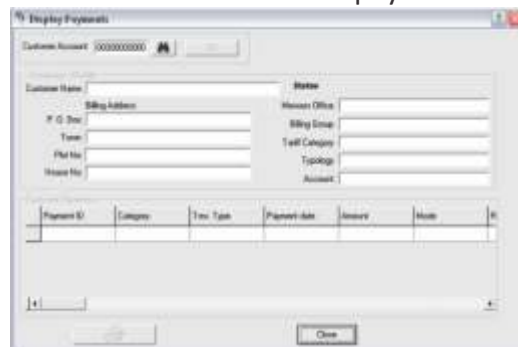


Figure 120: Display Payments Report Viewer

- Click the search icon button to obtain the customer account number.
- The customer details will be displayed at the customer profile section and the customer payments made will be displayed below the customer profile.
- Click the print icon button to preview the report before sending a copy to the printer.

6.9.2. *Print Payments*

This function is used to produce a report of a selected billing group. The report can be generated for a billing group or for a single customer in a selected billing group.

The layout is as follows:



Figure 121: Print Payments Report Builder

- Select the billing group/zone to print out the payment records.
- Select between the option of printing the entire billing group or a single customer. Single customers can be selected for printing only when a billing group is selected and not when the <All> option is chosen.
- Select the period (start date and end date), which will display the payments made in that period for the selected billing group.
- Click the print icon button to preview the report before producing a hard-copy.

6.9.3. Summary Daily Payments

This function is used to give a report on all the payments made on a daily basis. The report is displayed in a shortened way where information such as the Amounts paid and the processes customers go through (e.g. new contract fee, water consumption, deposits).

The layout:



Figure 122: Summary Daily Payments Report Builder

- Select the Company/Office that receives the payments.
- Select the period by selecting the Start and End date for the report.
- Select between the option of previewing or sending the report to a printer directly.
- Click the Print icon button to produce the report and close to exit.

6.9.4. *Detailed Daily Payments*

This function is used to provide reports that are detailed with information about the customer payments made per day. It contains the customer name, the payments made and the services paid for (Deposits, water consumption, new contract fee etc.)

The layout:



Figure 123: Detailed Daily Payment Report Builder

- Select the Company/Office that receives the payments.
- Select the report range period by selecting the Start and End date for the report.
- Select between the option of previewing or sending the report to a printer directly.
- Click the Print icon button to produce the report.

6.9.5. *Cashier Daily Payments*

This report is used to generate the payments received from a specified terminal assigned to a cashier for a day.

The layout:

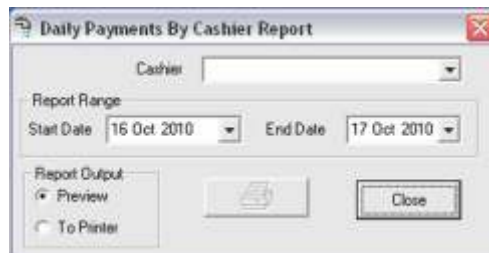


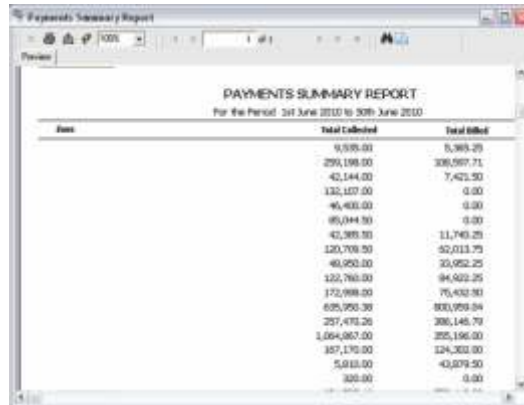
Figure 124: Daily Payments Received by a Cashier Report Builder

- Select the cashier's name from the drop down menu.
- Select the report range period that you would like to see.
- Select the report output by selecting to preview or to send a copy to the printer directly.
- Click the print icon button to produce the report or Close to end.

6.9.6. *Collection Efficiency*

This report is used to provide information for the finance department and the entire organization to make decisions. It contains the zones, total collected, total billed and collection efficiency (a percentile formula that compares the amount collected versus the amount billed over a specified period of time).

A sample report:



PAYMENTS SUMMARY REPORT
For the Period: 01 June 2010 to 30th June 2010

Date	Total Collected	Total Billed
01/06/10	5,395.00	5,395.25
02/06/10	299,198.00	300,597.71
03/06/10	42,144.00	7,421.50
04/06/10	132,117.00	0.00
05/06/10	95,400.00	0.00
06/06/10	85,044.00	0.00
07/06/10	41,385.00	11,740.25
08/06/10	120,700.50	52,013.75
09/06/10	40,950.00	30,952.25
10/06/10	122,760.00	94,502.25
11/06/10	172,898.00	76,432.50
12/06/10	605,950.30	803,959.04
13/06/10	257,470.20	300,145.75
14/06/10	1,064,867.00	355,196.00
15/06/10	157,170.00	124,303.00
16/06/10	5,010.00	43,079.50
17/06/10	300.00	0.00

Figure 125: Payment Summary Report

6.9.7. *Non-customer Receipts*

This is a function that is used to view reports on all the receipts issued by a specific cashier to non-customers after purchase of a service/good that the company is providing.

6.9.8. *Collection Report*

This is a function that displays reports for daily, monthly and other payments summary collection reports.

6.9.9. *Banking Report*

This function allows the user to access the daily report for the total amounts banked and in what form they were banked e.g. total cash, cheques etc.

This report comes from the update banking screen (refer to Chapter 6.8)

7. | Debt Management

7.1. DEBT MANAGEMENT GENERAL

The debt management function has sub-functions for facilitating follow-ups and management of customers with unpaid balances. Debt management menu is composed of the following sub- menus:

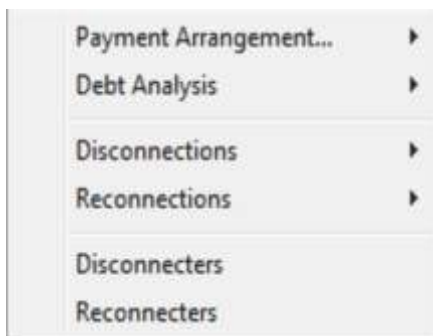


Figure 126: Debt Management Menu Items

7.2. PART PAYMENT ARRANGEMENT

This is a function that is crucial for customers who might have accumulated large balances or debts. The company and the customer can enter into an agreement on terms of payment of arrears over an agreeable time frame. There are various operations that can be done at this sub menu. These include:



Figure 127: Payment Arrangement Menu Items

7.2.1. Creating a New Payment Schedule

This is the process of enrolling a customer to the special service of clearing debt over a certain period of time. It takes the company and the customer to come up with an agreeable amount for payment and the period of time.

The layout:

Figure 128: New Payment Schedule Screen

- Click the search button to obtain the customer's account number
- The amount to be paid in total will be displayed for the customer.
- The MIS user should input the **number of instalments** and make sure that the water, sewer & other amounts have same value as first installment. Set the 1st installment date.
- Click Monthly or Weekly installments buttons for the system to calculate payments and due dates accordingly.
- Enter the amounts and due dates then click manual installments to register in the payment due list, until the account balance is cleared.
- Click the register list button to activate the arrangement for the customer.

7.2.2. *Print Arrangement*

The print function is used to produce a copy of the arrangement made by a customer and the company. It provides information such as how much has been paid and how much is remaining for them to complete the payment of the arrears. It also shows the due date that the customer ought to pay the stipulated amount. If this date is surpassed without payment, the customer is placed under the disconnection list.

The layout:



Figure 129: Print Payment Schedule Display Screen

- Click the search button to obtain the customer's Account Number.
- Select the report range period. The payment schedules will be displayed at the Payment Schedule Listing pane and the payment schedule details will be displayed.
- Click the print icon button to preview the customer's payment schedule and the progress made in paying the account balance.

7.2.3. *Modify Arrangement*

The modify function is used to change an already created payment schedule list for a customer. It can be used to delete or terminate a payment schedule for a customer.

The layout:



Figure 130: Modify Payment Schedule Screen

- Select the account number of the customer by clicking the search button and using the search criteria.
- Select the report range period as required, to show all the payment schedules available for the customer.
- These will be displayed at the pane right below the report range
- Select one of the payment schedules to delete. Click the delete button to complete the process.
- Select one of the payment schedule displayed to terminate it. Click the terminate button to complete the process.

7.2.4. *List Arrangements*

This function displays an inventory of all the customers in a billing group with a payment schedule. You can also view all the customers in the company with a payment arrangement.

The layout:

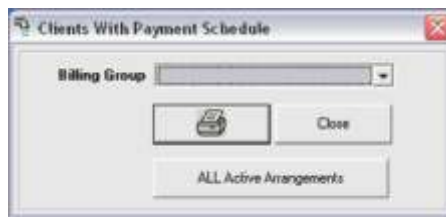


Figure 131: List Arrangements Report Builder Screen

- Select the billing group name to identify the number of clients with payment schedules.
- Click the print icon button to preview a report before the actual printing.
- Click **all active arrangements** button to view all customers who have not yet cleared their total arrangement amount in the zone.

7.3. DEBT ANALYSIS

This is a function that deals with determining the amount of arrears a customer has accumulated. It helps the company to identify customers who are not credit worthy. It contains a sub-menu know as Age Analysis.

7.3.1. *Age Analysis*

Age analysis is a function used to run through a billing group and or categories and identify any pending debts and displays overdue debts categorized by Zones/Billing Groups or by Customer Category.

The layout:



Figure 132: Age Analysis Report Builder

- Select the billing group name/ Customer Category to be examined.
- Select the Detailed or Summary option and order by desired criteria.
- Click the Display button to view the age of overdue debts in months (3 months, 4 months and 5 months over due etc.)
- To view the summary of all the billing group debts select the Summary option above the Close button and Click the Display button to view the report.
- Click the Close button to exit the window.

7.4. DISCONNECTIONS

This is a function that enables compilation of disconnection orders, capture of the disconnections effected on site and disconnection data analysis.

The following represents the sub-menus found in disconnections:

- > Compile Disconnections
- > Disconnection Capture
- > Disconnection Capture (Special)
- > SMS Disconnection notice
- > Print Notice/Letters

7.4.1. *Compile Disconnections*

This function allows the user to compile a list of customers who have overdue debts that require disconnection for defaulting in payment of bill(s).

The layout:



Figure 133: Disconnection Order Compilation Report

- Select the billing group to create a disconnection order list.
- To see a list with various selected criteria choose among the following filters (Minimum a/c balance, period of unpaid bill e.t.c.)
- Click Extract List button to view the list according to filters selected.
- One has the option of checking (✓) on the a/c due for disconnection to be included in the disconnection list. Click Print Disconnection List to produce a list for technicians to effect on ground.

7.4.2. *SMS Disconnection Notice*

This function allows the user to compile a list of customers who have overdue debts that require disconnection for defaulting in payment of bill(s) and proceed to send notices via Short Messaging System (SMS).

- Select the billing group to create a disconnection order list.
- To see a list with various selected criteria choose among the following filters (Minimum a/c balance, period of unpaid bill e.t.c.)
- Click Extract List button to view the list according to filters selected.
- Select the due disconnections and Click the send Via SMS button to queue the SMS notifications to send via a 3rd Party application.

7.4.3. Disconnection Capture

This facilitates the capture of the actual disconnections effected on the client's site. It includes the capturing of the meter reading and date disconnection was carried out.

The user selects the disconnection capture and is presented with a display with the billing groups who have customers with overdue debts that can be refined by searching for the account that has been disconnected.

Select the billing group name to view the customers available. The following screen will be displayed.

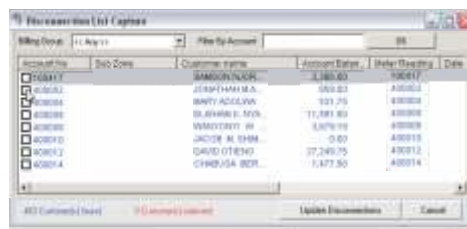


Figure 134: Disconnect Customer Selection Screen

This form will give the user the privilege to enter information about the disconnection process for a specific customer. The following form will be displayed when you click on a customer name.

Figure 135: Record Reading at Disconnection Screen

- This form captures the meter reading as at the time of disconnection.
- Enter the technician who disconnected the connection and the disconnection date.
- Select the disconnection mode and enter reading code if any.
- Select the method used to disconnect the meter and add remarks that are related to the disconnection and click Update button.

7.4.4. *Print Notices/Letters...*

This is a mail merge function used to print out letters to customers to inform them of disconnection orders detailing the balances and due date information. A template document with the intended message is placed on the machine used to send out (via email) or print out the notices.

The Layout:

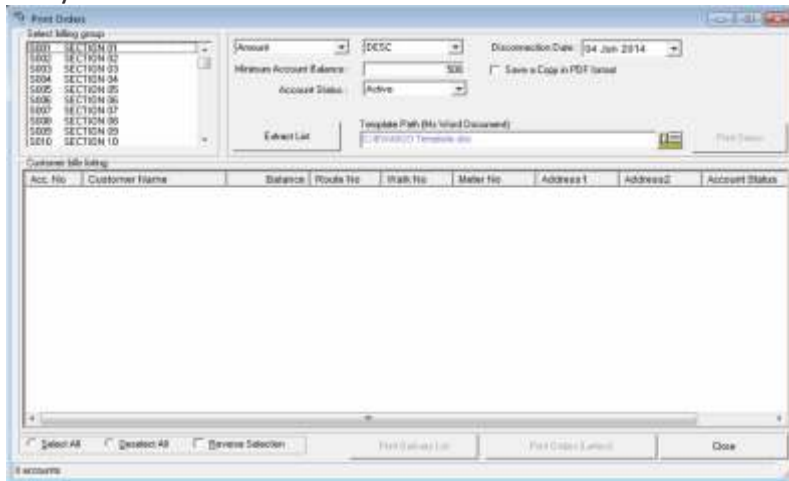


Figure 136: Print Notice Screen

- Select the billing group, order by amount / Account and set the minimum balance. You may also set the account status and due date required.
- Click the Extract list button to view the accounts that meet your set parameters. Once they appear. Select the desired accounts.
- Ensure that the path to the MS Word template is correct.
- Click Print Orders button to send the letters to printer
- Click the Print Delivery List button to print out a list of the accounts selected.

7.5. RECONNECTIONS

The function facilitates the recording of reconnection reading data, after it has been connected on site. A reconnection can be made only if a customer has paid all outstanding bills and reconnection fees. Note that you can only reconnect a disconnected customer.

7.5.1. *Reconnection Capture*

This function facilitates reconnections as explained above.

The layout:



Figure 137: Reconnection List Group Selector Screen

- Select the Billing group with customers that have been reconnected.
- Click the ok button to proceed. The following screen will appear:



Figure 138: Customer Reconnection List Selector Screen

- Click to check the customers reconnected. The following window opens:



Figure 139: Customer Reading Entry at Reconnection Screen

- Enter the required information on the above window and click the Update button. At the Customer Reconnection list selector screen select the Update Reconnections button to complete the process.

7.6. DISCONNECTORS

This is a function that facilitates the management of meter reader details that can carry out a disconnection on a client's site. It includes creating deleting and editing the details of a meter disconnection expert.

The layout:



Figure 140: Meter Disconnector Management Screen

- The compulsory fields to be filled include the ID, First Name and Other Names. To symbolize that a field is blank, use a hyphen symbol in the provided fields (-).
- Click the Create button to register the employee to the system.
- To modify an already existing record, Click the Search button and double-click the meter reader name to edit details.
- Change the information where necessary on the above form and click the Modify Button.
- To delete a meter disconnector, click the search Button, obtain the name by double clicking it from the record. Click the Delete button which can be found at the bottom of the form adjacent to the Modify Button.

7.7. RECONNECTORS

This function is used to register, de-register and edit artisans who are able to carry-out a reconnection procedure on a client's connection.

The layout:

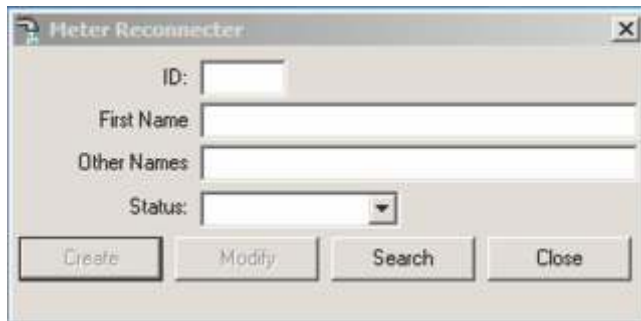


Figure 141: Meter Reconnector Management Screen

- Enter the following compulsory fields to create a meter reader: the ID, First Name and Other Names. The rest of the details can be replaced with a hyphen (-) to show that the detail is not available or applicable.
- Click the Create button to register a meter reader with the privilege.
- To modify an already existing record, Click the Search button and double-click the meter reader name to edit details about.
- Change the information where necessary on the above form and click the Modify Button.
- To delete a meter reader who can conduct disconnections, click the search Button, obtain the name of the meter reader by double clicking on the record. Click the Delete button which can be found at the bottom of the form adjacent to the Modify Button.

8. | Meter Management

8.1. METER MANAGEMENT GENERAL

This function is created for the purpose of administering meters owned by the company. There are various functions that can be conducted at this menu item: They include specifying meter types, maintaining meters, modifying meter details, facilitating and managing meter movements and producing reports about the meters. The following is the layout of the menu items.

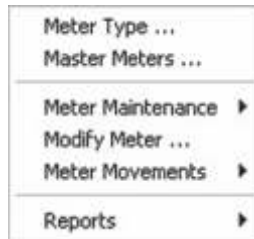


Figure 142: Meter Management Menu Items

8.2. METER TYPE

This is a function dedicated to managing the types of meters in the company and the system. The following are various actions that can be performed on the meter in the system: Creation, deletion and Editing meter type details. The meter type is a category that tends to describe the features of meters from a manufacturer.

The layout:

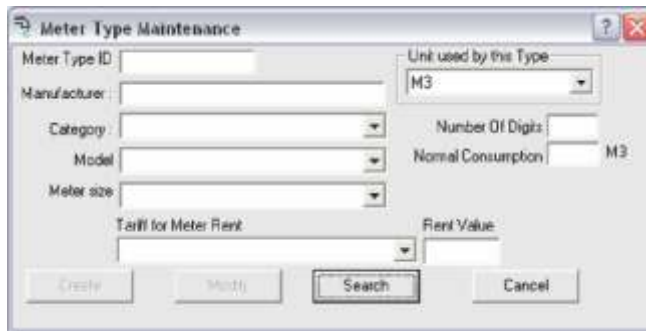


Figure 143: Meter Type Maintenance Screen

To create a new meter type:

- Enter the compulsory fields in the form.
- Click the Create button to register the meter type in the system.

To modify Meter Type details:

- Click the search to obtain an existing Meter Type.
- Make the necessary changes to the meter type details.
- Click **Modify** Button to save the changes.
- Click **Cancel** button to close the form.

To delete a Meter Type:

- Use the search to obtain a Meter Type.
- Click the **Delete** Button.
- Click **Cancel** button to close.

8.3. MASTER METERS

This is a function dedicated to managing the meters that handle large volumes of water used by the company as feeder meters to a zone or area. These meters are designed to show the company how much water is being transmitted to the area or zone. Additional features such as the GIS location data can be captured

The layout:

Figure 144: Master Meter Maintenance Screen

To create a new master meter:

- Enter the necessary data in the fields provided in the form.
- Click the Create button to register the master meter in the system.

To modify Master Meter details:

- Click the search to obtain an existing Master Meter.
- Make the necessary changes to the details.
- Click **Modify** Button to save the changes.
- Click **Cancel** button to close the form.

To delete a Master Meter:

- Use the search to obtain a Master Meter.
- Click the **Delete** Button to complete the process.
- Click **Cancel** button to close.

8.4. METER MAINTENANCE

This sub-menu item is used to manage meters found either in the company premises or in a client's site. It contains two modules which include registering new meters and modifying a meter status in the system.

The sub-menus include:



Figure 145: Meter Maintenance Menu Items

8.4.1. *Registering a New Meter*

The function is used to capture the details of a new meter (issued from the warehouse or manufacturer for the first time). This action records the meter into the system which renders it useful until it reaches its disposal age.

The layout:

A screenshot of the 'Register Meter' dialog box. It contains a checkbox for 'Register Meters in Batch', a text field for 'Meter number', a dropdown for 'Meter type', a dropdown for 'Meter Condition', a dropdown for 'Comments', and a date field for 'Date' showing '05 Nov 2009'. At the bottom are 'Create' and 'Cancel' buttons.

Figure 146: Register Meter Screen

- Check the option “Register meters in batch” if you would like to register sequential meter numbers in bulk from a supplier.
- Enter the Meter number, Meter type, Condition and any comments. Enter the date of registration for the meter.
- Click the create button to capture the meter details to the system.

8.4.2. *Modify Meter Status*

This function is used to change particular details about an existing and registered meter. You can change the meter readings, the meter status and date of status change, and any comments.

Scenarios where meter status can change include; the repair of a meter which was initially broken or rendered unusable, installation of a repaired meter on a client's site, a broken meter being moved to the warehouse for repair e.t.c.

The layout:



Figure 147: Modify Meter Screen

- Click the search icon button to obtain the meter number of the meter to modify the status.
- The meter details and the customer currently using the meter will be displayed. If the meter is at the store then no customer profile will be displayed.
- The only area that can be edited is the “new values” section which contains the Meter reading, the status date, meter status and comments.
- When done with editing these fields, click the update button to complete the process.

8.5. MODIFY METER

This function is used to change the meter details which include meter number and meter type.

In this function the meter details can be changed even when the meter is installed on the customer's premises. It is mostly used when the details about a meter are clarified.

The layout:



Figure 148: Meter Maintenance Menu Items

- Click the search button to obtain the meter number of the meter to modify.
- The meter details and the customer currently using the meter will be displayed. If the meter is at the store then no customer profile will be displayed.
- The only area that can be edited is the “new values” section which contains the Meter Number and meter type.
- When done with editing these fields, click the update button to complete the process.

8.6. METER MOVEMENTS

This function is set for the user to record and manage the details of various meter events; these include the exact locations of the meters and tracking of the meters as they circulate within the company or client's site, the history of the meters can be tracked too.

The following is contained in this sub-menu:



Figure 149: Meter Movements Menu Items

8.6.1. Meter Installation

This is a function used to reflect the actual fitting of the meter at the client's site. It is the last activity to carry out for a new customer to activate their account fully. This function records the details about the meter connection made at the customer's site. The meter must have been registered into the system before this process.

The layout:

Figure 150: Meter installation Screen

- Select the first search button to obtain the meter number (Refer to meter search) and the Customer Account (Refer to customer search).
- The meter details will be displayed separately and the customer profile on its own too. Fill in the remaining fields and click Install.

8.6.2. *Meter Removal*

This function is used to reflect the removal of a meter from the client's site back to the company in the system. It could happen if a meter becomes faulty or the customer's contract is terminated or a disconnection is effected. The meter must have been installed on a client's site before it can be removed.

The layout:

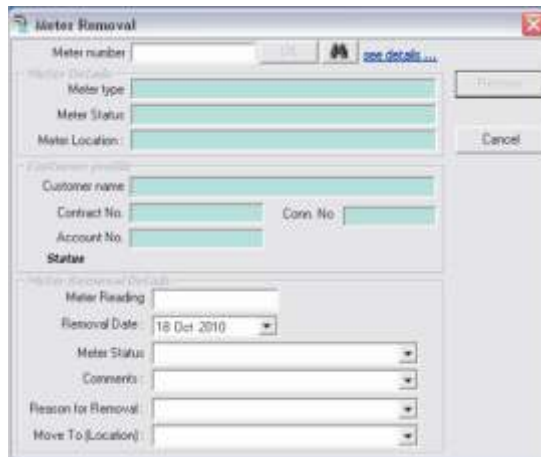


Figure 151: Meter Removal Screen

- Click the search icon button to obtain the meter number.
- The meter details and the customer profile will be displayed.
- Enter the required details in the Meter Removal Details section. (Meter reading, Removal date, the meter status, comments, the reason for removal and if the meter has been moved to another location)
- Click the Remove button to reflect the change in the customer's account

8.6.3. *Inter-section Transfer*

This function facilitates the movement of a meter within the departments in the company. It is used by the user to move the location once it has been effected. The meter could be moving within departments in the company for purposes of repair and maintenance. The function can be used by the customer service to explain to the client the status of his meter.

The layout:



Figure 152: Inter-section Meter Transfer Screen

- Select the search button to obtain the meter number.
- The meter details and the customer profile will be displayed on their respective sections.
- Enter the details of the New Meter Location to signify the transfer of the meter.
- Click the Transfer button to complete the process.

8.7. REPORTS

This is a function that produces statements showing information about meters. The following is what is contained in the report sub-menu.

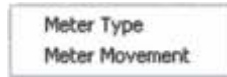


Figure 153: Meter Management Menu Items

8.7.1. *Meter Type Listing*

This function gives a report of all the meter types found in the system/ company. It provides the details of the meter type in the same report.

8.7.2. *Meter Movement Report*

This function gives a report of a meter's activity that has taken place over a specific period of time. The report only allows the user to produce a report for one meter.

The following is the layout:

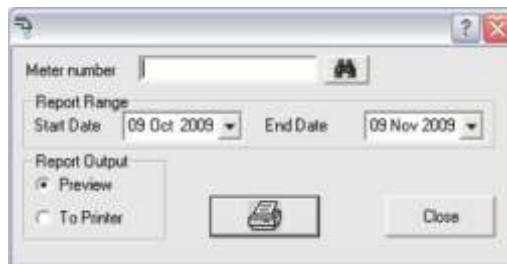


Figure 154: Meter Movement Report Builder Screen

- Click the search icon button to obtain the meter number in question.
- Select the report range dates which include the Start Date and the End date to view the movement within the specified dates.
- Select the option of previewing or sending the report to the printer.
- Click the print icon button to execute the selected options.

9. | REPORTS

9.1. REPORTS OVERVIEW

This is a menu that contains management functions that are used for compilation, display and printing of reports and management summaries. The report maintenance module is used by the administrators, managers and other system users. The following is contained in this menu:



Figure 155: System Reports Menu Items

9.2. REPORT FINDER

This function is used to store and locate any report found in the system. The layout is as follows:

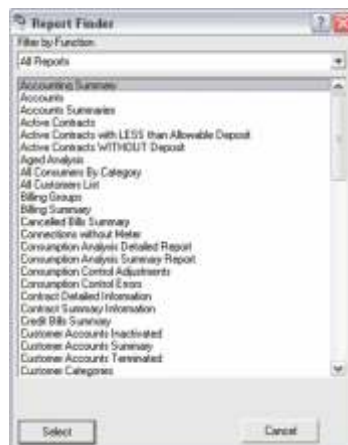


Figure 156: Report Finder Selector Screen

- Scroll down until you find the right report you would like to see then double-click to load or click the report name and click the select button below.
- When the report appears, select the required information such as the report duration, the office affected and the billing group name. Click the print button to activate a preview or to make a hard copy. The information to be entered varies for each form.

10. | System and Administration

10.1. ADMINISTRATION GENERAL

This function has sub-functions for administering the system. This section of the system is to be used by the system administrator. The sub-functions include tariff management, parameters management, security, backup and restore, database management and reports.

The following is what is contained in the menu:



Figure 157: Administration Menu Items

10.2. TARIFF MANAGEMENT

This function is used by the administrator to modify, duplicate and delete the levy used by the company on consumer categories. This levy determines the amount a customer category will pay periodically.

10.2.1. Modify Tariff

The layout is as follows:

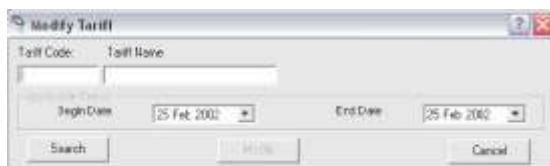


Figure 158: Tariff Modification Screen

- Click the search button to obtain a tariff from the system
- The details of the tariff chosen will be displayed (This includes the tariff category and the tariff line)

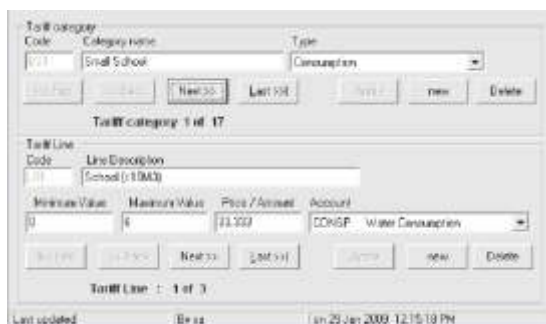


Figure 159: Detailed Tariff Information Screen

- You can create a new type for a tariff or delete it.
- Click the next button to move to different categories and edit them.
- In the Tariff Line section, more details can be edited. These include the minimum and the maximum value, the price for that value and the account affected. In this section a New Account can be created.
- Once you have finished editing click the Modify Button to save the changes.

10.2.2. Duplicate Tariff

It is similar to the modify form. The difference is that in this form the loaded tariff is given a new name and code, possibly, the start and end date for the tariff applicable date is also given and then you can create new tariffs leaving the older one unedited.

The layout is as follows:



Figure 160: Duplicate Tariff Screen

- Click the search icon button and select the tariff you would like to duplicate that is already existing in the database.
- Click the duplicate button to make a copy of the tariff.

A new section within the window will be displayed:

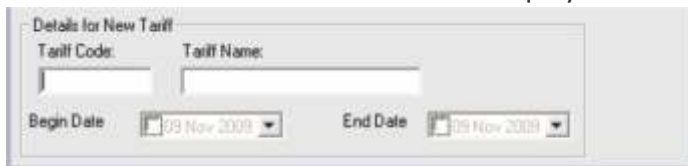


Figure 161: Duplicate Entry of New Tariff Screen

- Enter the Tariff code and the tariff name.
- Select the begin date and the end date within which the tariff will be active.
- Click the ok button to complete the task.

10.3. PARAMETERS MANAGEMENT

This has a set of sub-functions that facilitate the capture and modification of the operational company information by the system. The administrator is able to search for, display and modify the different modules of information managed here.

The content of this sub-menu includes:

10.3.1. *System Settings*

This function is used to facilitate the entry of various default system settings used for bills and reports. This information can be edited and printed.

These include tabs with the following information:

Information about the Company Profile:

- Company name - Company Location
- Company logo - Institutional message
- Company Addresses - Seasonal message

Information about the Institutional Structure

- Company Organisation Structure
- Physical Location Structure

Information about default Settings

- The address structure - Conversion rates
- Default currency code
- Unit of measurement - Units of volume

Information about the Billing system

- Max consumption gap - Consumption per period above normal
- Max consumption gap - Grace period for journal payment below normal
- Max value (bill line) - Current tariff to apply
- Min amount (disconnection)
- Grace period for payment - Disconnection message



Figure 162: Parameter Management Menu Items

10.3.2. *Accounts*

This facilitates the management of the various accounts found in the company for clients, cash, revenue, expenses e.t.c. to be used in the financial interface.



Figure 163: Account Maintenance Screen

To create a new account:

- Select the account type, enter the account code and the name of the account. Select the validity of the accounting period (Opening and Closing dates for a financial period)
- Click the create button to register that account.

To modify an existing account:

- Click the search button and obtain an account name that you are interested in.
- The details will appear on the form, change the necessary fields and click the modify button to complete the process.

To delete an existing account:

- Click the search button and select the account name to delete from the system.
- The details will be displayed on the form to counter-check that it is the right account.
- Click the delete button to remove the account from the system.

9.3.3. *Billing Group*

This facilitates the management (creation, modification and deletion) of clients put in a cluster for purposes of issuing financial statements according to the company's method of billing.

The layout:

Figure 164: Billing Group Management Screen

Creating a new billing group:

- Enter a group code and the Group name.
- Select the office responsible for handling the billing group and from the options given select the billing method.
- Click the Create button to create a billing group.

Modifying a Billing group:

- Click the search button to obtain an already existing billing group.
- The details about the billing group are displayed on the form.
- Change the required details about the billing group.
- Click the Modify button to save the changes.

Deleting a Billing group:

- Click the search button to obtain an already existing billing group.
- The details about the billing group are displayed on the form.
- Click the Delete button to remove the billing group from the system.

10.3.4. Customer Typology

This function facilitates the management (creation, modification and deletion) of different type of customers in the region.

The layout:

Figure 165: Customer Typology Management Screen

To create a new customer typology:

- Enter the code and the name of the customer typology.
- Enter and select the normal consumption capacity (m³ or Gal) and maximum value for a bill line for a customer type.
- Select the type of premises to determine the deposit amount for different clients.
- Click create button to register the new customer typology in the system

To modify an existing customer typology:

- Click the search button to obtain an existing customer typology.
- Edit the required fields to the required values.
- Click the modify button to save the changes made.

To delete an existing customer typology:

- Click the search button to obtain the name of the customer typology.
- Click the delete button on the form to remove the customer typology from the system.

10.3.5. *Geographical Location*

This function facilitates the management (Creation, modification and deletion) of location information to be used in the system.

The layout:



Figure 166: Geographical Location Management Screen

To create a new geographical location:

- You must type the short code for the location (E.g. Nairobi as NBI)
- Enter the location name in full

- Click the create button to store the new location.
- To move to the next tab, first select the newly created location at the code section on the form.
- Click the next tab to enter the required details (Type in the Meter Code and the name)
- Click the third and final tab to enter the necessary information.
- Click the create button to finalize the registration of the location.

To modify an existing location:

- Select the short code that is already in the system
- Click the subsequent tabs and change the necessary fields.
- Click the modify button to save the changes on the system.

To delete an existing location:

- Select a code on the first tab and for the subsequent tabs, select the specification of the location area you would like to delete.
- Click the delete button to remove the selected location from the system.

10.3.6. *Company Organization*

This facilitates the management (creation, modification and deletion) of the hierarchical nature of the organization that portrays the organizational structure to determine the user rights for employees in the company.

The layout:

Figure 167: Company Organization Management Screen

- Type in the code for the company and the full name to create the HQ/Board office.
- Select the HQ's code to display the full name just below the code. Select the next tab and enter the scheme information.
- Select the next tab to select the office name.

- You can change the existing company organisation structure by editing the available fields and clicking the modify button to save changes.

10.3.7. *Disconnection Profile*

This function facilitates the management (creation, modification and deletion) of a disconnection profile that will be used in the system to enforce the grace period before a disconnection list can be prepared.

The layout:



Figure 168: Customer Disconnection Profile Screen

To create a new disconnection profile:

- Enter the profile code and the name of the profile
- Enter the grace period to be applied.
- Click the create button

To modify an already existing disconnection profile:

- Click the search button and select the name of the disconnection profile that you would like to change details about.
- Change the desired field information on the form and click the modify button to save the changes.

To delete an existing disconnection profile:

- Click the search button and obtain the name of the disconnection profile to be removed from the system.
- Click the delete button to remove the disconnection profile.

10.3.8. Disconnection Mode

This function facilitates the management (creation, modification and deletion) of the method used in disconnecting meters from client's site. The layout:



Figure 169: Disconnection Mode Creation Screen

To create a new disconnection mode:

- Enter the code and the description that matches the code.
- Click the create button

To modify a disconnection mode

- Click the search button to obtain an existing disconnection mode.
- Change the necessary information on the fields and click the modify button to save the changes.

To delete a disconnection mode:

- Click the search button to obtain an existing disconnection mode.
- Click the delete button to remove the mode from the system.

10.3.9. Customer Tariff Category

This function facilitates the management (creation, modification and deletion) of a customer levy for various categories of customers to be applied for a service offered by the company.

The following is the layout:



Figure 170: Customer Tariff Category Screen

To create a new customer Tariff Category:

- Enter the Category code and the category name, account and tariff to be applied and click the Create button.

To Modify a Tariff Category:

- Click the search button to obtain an existing tariff category from the system
- Change the required fields in the form
- Click the Modify button to save the changes to the system

To Delete a Tariff Category:

- Click the search button and select the name of the tariff category
- Click the Delete button to remove the tariff category from the system.

10.3.10. Look-up Table

This function facilitates the display of various tables used by the system. Since the information displayed here is very sensitive, the user is not allowed to change anything.

The layout:



Figure 171: Look - Up Table Management Screen

- To view information about tables and their dependencies in the system, select the table class and select the options in the line code field.
- The description and comments about the table will be displayed.

10.3.11. Payment Category

This function facilitates the management (creation, modification and deletion) of the cluster in which customers have common factors considered e.g. paying for a particular service offered by the company.

The Layout:

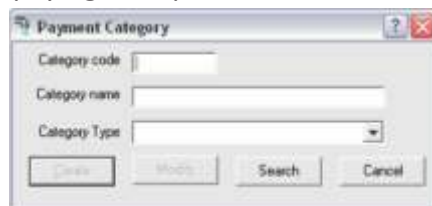


Figure 172 : Payment Category Management Screen

To create a new payment category:

- Enter the category code and category name
- Select the category type the new payment category will fall under.
- Click the Create button.

To modify a payment category:

- Click the search button and select existing payment category.
- Edit the required fields and click the Modify button

To delete a payment category:

- Click the search button to obtain existing payment category and select it.
- Click the delete button to remove it from the system.

10.3.12. Accounting Period

This function is used to manage (create, delete and modify) financial periods that determine the financial years in the company.

The Layout:



Figure 173: Accounting Period Management Screen

- Select the month and the year that you would like to create an accounting period for.
- Select the validity time for that accounting period (this includes the opening and closing date for the period).
- Click the create button.

10.3.13. *Modifying Account Period*

This function manages the change of status on already created accounting periods in the system.

The layout:



Figure 174: Modify Accounting Period Screen

- In the drop down menu select the existing accounting period.
- In the period status section, select the option that is appropriate.
- Click OK button to activate the option chosen.

10.3.14. *Reading Code*

This function facilitates the creation, deletion and modification of a reading code that informs the system user of the reading circumstances encountered at a client's site.

The layout:



Figure 175: Reading Code Management Screen

To create a new reading code:

- Enter the code and the full name of the reading code.
- Select the reading quality to be used in this code.
- Select the option of displaying the anomalies in a special anomaly report.
- Click the create button to make a new reading code

To modify a reading code:

- Click the search button to obtain existing reading codes in the system. Select the one you want to modify and click ok
- Change the required fields and click the modify button to save changes.

To delete a reading code:

- Click the search button and obtain the reading code.
- Click the delete button to remove the reading code from the system.

10.3.15. Complaints

This function facilitates the creation of modules that will help in the collection of customer's views and grievances to help improve the customer experience used by the customer relations department in the company.

It contains the following menu items:



Figure 176: Complaints Menu Items

10.3.15.1. Complaints Code

This facilitates the management of codes that are used to describe the nature of a complaint in the system.

The layout:



Figure 177: Complaint Codes Management Screen

- To create a new complaint code, enter the code and the description in the required field.
- Select the nature of the complaint (e.g. administrative, technical etc.)
- Click the create button to register a new complaint code.
- To modify an existing complaint code, click the search button and select the required complaint code, edit it and click the modify button to save the changes.
- To delete a complaint code, click the search button and obtain the complaint code to remove from the system then click the Delete button.

10.3.15.2. *Department Code*

This facilitates the management of codes that represent departments found in the organization.

The Layout:

Figure 178: Department Management Screen

- To create a department code, enter the short name of the code and the full name of the code in the Name field.
- Select the name of the leader that heads that department.
- Click the create button to register the department.
- To modify an existing department code, click the search button and obtain the name of the department in question. Edit/change the necessary fields and click the modify button to save the changes.
- To delete an existing department code, click the search button and obtain the department code or name. Click the delete button to remove the entry.

10.3.15.3. *Posts*

This facilitates the management of company positions that are specialized for employees of the company to take up.

The layout:

Figure 179: Post Management Screen

- To create a new post, select the department that will manage that post (e.g. Customer relations, technical). Enter the code name and the description.
- Check (✓) if the holder of the post can re-activate a resolved complaint.
- Click the create button.

10.3.15.4. *Complaints Priority*

This facilitates the management of complaints put in by customers or employees. It creates, modifies and deletes the priority list of options for a solution.

The layout:



Figure 180: Compliant Priority Management Screen

- Enter the name of the priority and the description of the name then click the create button.
- To modify an existing priority type, click the search button and obtain a priority type from the list provided. Edit the description or the name of the priority type and click the modify button to save the changes.
- To delete a priority type, click the search button and obtain the intended priority type. Click the delete button to remove the complaint priority type from the system.

10.3.15.5. *Complaints Status*

This facilitates the management of complaints by creating modifying or deleting the status name of the progress of the complaint put in by the customer or employee in the system.

The layout:

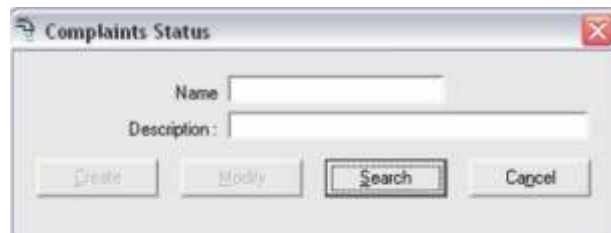


Figure 181: Complaint Status Management Screen

- Enter the name of complaint status and the description. Click the create button to register a new complaint status.
- To modify a status, click the search button and obtain the status to change. Edit the required fields and click the Modify button.
- To delete a status, click the search button, select a status and click the delete button.

10.3.15.6. *Employees*

This facilitates the management (creation, modification or deletion) of the employees in the company via the MIS software.

The layout:

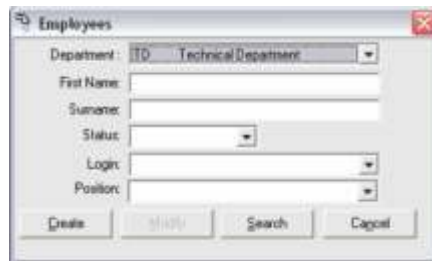


Figure 182 : Employee Management Screen

- Select the department the employee will be working in.
- Enter the first name and the surname of the employee.
- Select the status of the employee (Active/ inactive)
- Select the login name for the employee to the system and the position he/ she takes in the department.
- Click the create button to store the new employee.
- To edit the details of an existing employee, click the search button, obtain the name of the employee and select it. Change the necessary fields and click the modify button to save the changes.
- To delete an employee, click the search button select the user and click the delete button.

10.3.15.7. *Complaints Category*

This facilitates the management (creation, modification or deletion) of a cluster, represented by codes, of complaints generated internally (employees) or externally (clients).

The layout:



Figure 183: Complaint Category Management Screen

- Enter the short code for the category and its full description in the subsequent fields.
- Click the Create button to save the new complaint category.
- To modify a category, click the search button and obtain the code of the complaint category you want to edit.
- Select it and change where necessary, click the modify button to save the changes.
- To delete a category, click the search button and obtain the complaint category to remove. Select it and click the Delete button to remove from the system.

10.4. SECURITY

This function facilitates enforcement of options that control the accessibility of modules or functions in the system and managing the users and the user groups found in the system. The following is contained in the sub-menu:



Figure 184: Security Menu Items

10.4.1. User Groups

This is a function that manages the creation of clusters that contain several users with similar attributes. These clusters are created edited and/or deleted on this form.

The layout:



Figure 185: User Group Management Screen

- To create a new user group, enter the code group and the name of the group. Select the group role for the users in the group. Click create button to save the newly created user group.
- To modify a user group, click the search button obtain the user group to edit. Select it, change the fields that are necessary and click the modify button.
- To delete a user group, click the search button and obtain the user group. Click the delete button.

10.4.2. *Set Group Permissions*

This function is used to assign system users with rights to access and use certain functions in the system. The system administrator is responsible for assigning the user group's permissions. It contains a list of all the menus and submenus which the system admin can check (✓) to activate a function for a user group.

The layout:



Figure 186: Permission Selection for User groups Management Screen

- Select a user group to apply the permissions to at the drop-down menu. Go through every tab ticking the appropriate functions to activate or un-check to de-activate functions for user-groups.
- Click the search button and obtain a user group (e.g. administrators) check or uncheck the functions to activate or de-activate the click the Modify button to save the changes.
- To update all groups with certain functionality, locate the functionality and check it/un-check it. Click the update all groups button at the top right side of the form.
- To delete the permissions for a group, click the search button and obtain the user group, the permissions will be displayed. Click the delete button to remove any permission assigned.

10.4.3. *Users*

This function is used to manage the system users by creating users and modifying the status.

10.4.3.1. *Create Users*

This facility allows for the admin to register new system users.



Figure 187: Create New User Screen

- Select the user group the new user will be affiliated with.
- Enter the full name of the user, the user name to be used to login and the password. Confirm the password typed in the password field.
- Click the create button to register the user to the system.

10.4.3.2. *Change User Status*

This function allows for the administrator to modify the status of the system users.

The layout:

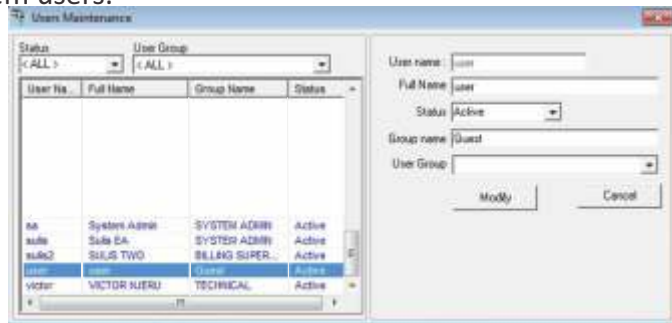


Figure 188: Change User Status Screen

- Click the specific user on the left section as shown above.
- The information about the user will be displayed on the right side of the form.
- You can change the status from active to inactive or vice-versa. Click the User Group drop-down to select a new group from the list, for the user.
- Click the modify button to save the changes made.

10.4.3.3. *Financial Adjustment*

This function allows for the administrator to assign system users permissions to prepare, recommend, approve and/or authorize financial adjustments in Majics.

The layout:

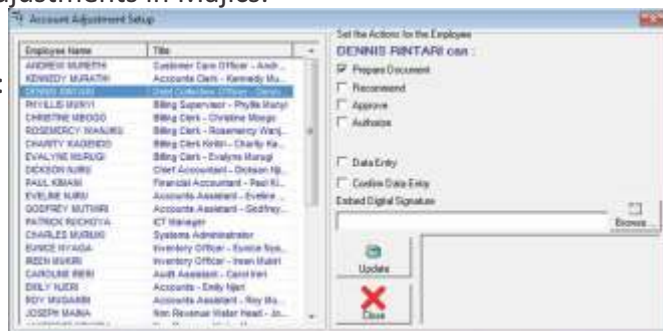


Figure 189: Adjustment Setup Screen

10.5. BACKUP AND RESTORE DATA

This functionality is available for the administrator to help them manage the storing of a copy of the database remotely or on recordable media (CD, DVD) and repairing it by restoring it when there is a system failure or corruption.

10.5.1. Backup

This facility is used by the administrator to make a copy of the current database used by the system in a proper state to an external media (Flash drives, CD/DVD media etc). It is a step by step process guiding the user.

The layout:



Figure 190: Backup Wizard Initiator Screen

Follow the wizard to complete the procedure of backing up data.

10.5.2. Restore

This facilitates the repair of the database to a stable state before a system error occurred or due to unforeseen circumstances. It is a step by step procedure guided a wizard.

The layout:



Figure 191: Restore Wizard Initiator Screen

10.6. DATABASE MANAGEMENT

This facilitates the administration of the system database to ensure there is no inconsistency (incoherency) of the various redundancies that can occur in the database. This is carried out periodically (e.g. quarterly or half-yearly). It has sub-functions for the various system checks. The function also has a sub function for compacting and/or de-fragmenting the database. It contains the following:

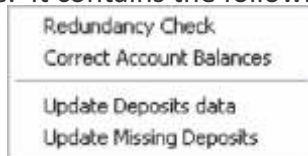


Figure 192: Database Management Menu Items

10.6.1. Redundancy Check

This facility assists the administrator in ensuring that there are no errors in modules that are crucial in the system. These include the bills, balances, payments and Metered and unmetered customers. The layout is as follows:

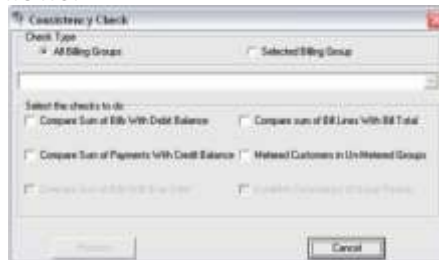


Figure 193: Redundancy Check Processing Screen

Select the option of examining all bills or a selected billing group to examine.

Check (✓) the system options you would like to perform and click process.

The following is the progress it shows



Figure 194: Display Progress Status Screen

10.6.2. *Correct Account Balances*

This module is used to fix wrong client account balances. It calculates the sum of a customer's bill and payments, compares them and reports any anomalies found. It allows for the administrator to effect the changes of the customer account to correct any errors in the database.

The layout:

Figure 195: Correct Account Balance Screen

- Select the search engine icon to obtain the account number
- The customer profile and account balance is displayed. Updates to this account can be made here by debiting or crediting the balances then clicking the save button to store the new balances.
- If the customer had a payment arrangement then click the Arrangement update button to make the necessary payments.
- When all is balanced click the correct account balance button to effect the customer account.

10.6.3. *Update Deposit Data*

This module is used to revise the deposits transferred from the old system. The administrator can only correct or change the deposit transferred from the old system. Deposits keyed in the system cannot be corrected by this module.

The layout:

Figure 196: Update Deposit Data Screen

- Click the search icon button to obtain the customer account no.
- The customer profile will be displayed for viewing and the existing deposit details will be displayed too. Enter the correct deposit details and click the update button to correct the account deposits made.

10.6.4. Update Missing Deposit

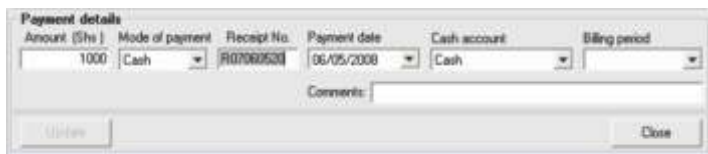
This function is used to correct customer accounts that do not reflect payment deposits initially made by the client, but the deposits were not updated in the system.

The layout:

Figure 197: Update Missing Deposit Screen

- Click the search icon button to obtain the customer account in question.

- The customer profile is displayed to certify that it is the same customer.
- An additional portion is displayed with field options such as:



The image shows a 'Payment details' window with the following fields and values:

Amount (Shs)	Mode of payment	Receipt No.	Payment date	Cash account	Billing period
1000	Cash	R07060530	06/05/2008	Cash	

Below the fields is a 'Comments:' text area. At the bottom are 'Update' and 'Close' buttons.

Figure 198: Payment Details Screen

- Enter the required data in the necessary fields to record the missing deposit.
- Click the update button to complete the process.

10.7. REPORTS

This facilitates the display of different system reports. These reports can be printed out in hard-copy or saved as softcopy files in the computer system. Some of these reports can be generated from system functions menu or sub-menus.

It contains the following reports:



Figure 199: Report Menu Items

10.7.1. System Settings

This report gives information about the system and all its settings (Refer to Parameter Management> System Settings)

Company profile

Company Name: [Text Box]

Company Address: [Text Box]

Location: [Text Box]

Institutional Message: [Text Box]
Help us serve you better by promptly reporting any complaint regarding your connection to our customer relations desk immediately, preferably in writing. You can request for your complaint reference number after reporting.

Seasonal Message: [Text Box]
Charges should be made payable to "Mera Water and Sewerage Services". The Due Date is for the monthly bill. For arrears, in arrears and arrears.

Disconnection Message: [Text Box]
A fee of RM. 600 to chargeable before reconnection.

Company Structure

Company Organization

Name of Level 1: [Text Box]

Name of Level 2: [Text Box]

Name of Level 3: [Text Box]

Physical Location

Name of Level 1: [Text Box]

Name of Level 2: [Text Box]

Name of Level 3: [Text Box]

Default Settings

Address title

Address 1: [Text Box]

Address 2: [Text Box]

Address 3: [Text Box]

Address 4: [Text Box]

Default Address

[Text Box]

Figure 200: System Report Layout Screen

Note: To view all the other reports, click on the report name that you would like to view from the sub-menu. It will display the report for printing, perusal or storing on the computer.

11. | Interfaces

11.1. INTERFACE GENERAL

Interfaces allow interaction between independent software created by different manufacturers. These interfaces help to link and share data between Majics MIS and other propriety software.

Available interfaces include:

11.2. FINANCIAL INTERFACE

This interface will interlink Majics MIS and propriety financial software that will require financial data from the MIS. Eg. SAGE Pastel

To use the interface, click the financial interface on the menu and follow the instructions on screen to obtain required results.

11.3. M-PESA INTERFACE

This feature is available for administrators for updating payments to the MIS, made by M-Pesa Service. A file containing the payment data is downloaded from the telecommunication service provider and the file uploaded to Majics.

To use the interface, click on the Mpesa interface on the menu and follow the instructions to update the system.

11.4. BANK INTERFACE

This feature is available for administrators. It updates customer account balances of the paid amounts, by receiving a file made available by the bank.

To use it click on the bank interface menu and follow the instructions to update the system.

11.5. GIS INTERFACE

This feature is used to export data to specialized geographical Information systems. Eg. ArchGIS. Data can be loaded from Majics and exported to a Comma Separated file (CSV). This file can be imported into a GIS software and analysis on the data carried out.

To initiate, click on the GIS interface menu and follow the instructions.

11.6. SMS PERIODICAL BILLS

The interface is used by the administrator to send monthly generated bills which have been ordered by zones/routes, to the respective customers via SMS. Once billing process is complete, the routes appear in the window used for transmission. Check the boxes of the billed zones/routes and click the send via sms button. Only the customers who have provided the correct phone number and set their delivery option as "SMS" will receive the bills using this feature. An option to send one bill to a specified phone number is provided.

12. | Search Engines

12.1. SEARCH ENGINES GENERAL

This is a tool that is found in various parts of the system to obtain records already found in the database. It allows the user to retrieve the records required by providing the system with known knowledge about a customer which is part of the search criteria in some cases. This will give the exact record being looked for or a list of possible records the system user might be looking for.

The following are the search engines found in the system under different functions.

12.2. CUSTOMER SEARCH ENGINE

This is a feature that is accessed by the system when a request for a customer account or details of a customer is made by a system user.

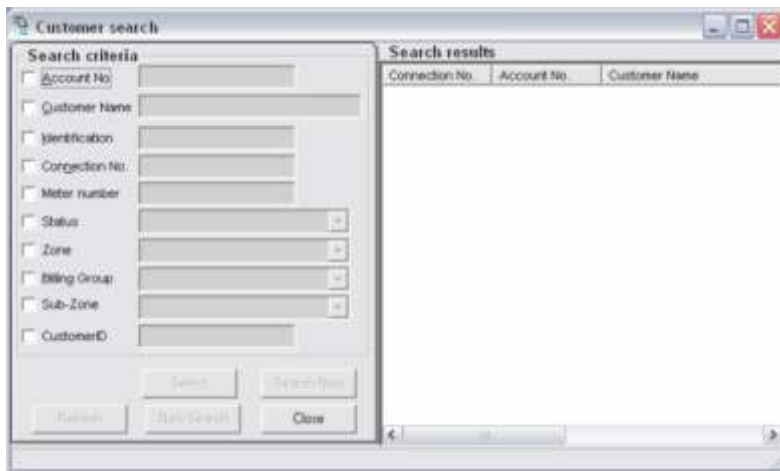


Figure 201: The Customer Search Engine Screen

To obtain a customer you need to select the check box which will be activated for you to enter the information about a customer you have so as to provide the closest result. This is known as the search criteria. You can enter a string or select a search option depending on the criteria you are using. For example, when you click the search criteria customer Name, you can enter the first or both names of the customer e.g. Papa or Papa Omusakuru. Click the Search Now button and the system will present you with all the entries in the database that have the First name as Papa. You could also choose to refine your search further by selecting another search criterion. For example, the criterion “Status” being an active account. This will imply that the system will provide all entries with the first name as Papa, in addition, it will only provide active contracts. To begin the process again without closing the window click New Search button.

12.3. COMPLAINT SEARCH ENGINE

This is a function that is used in the system to obtain various complaints that have been presented by a customer to the company over time. It also contains the search criteria items which have to be checked (✓) to be activated for the MIS user to use.

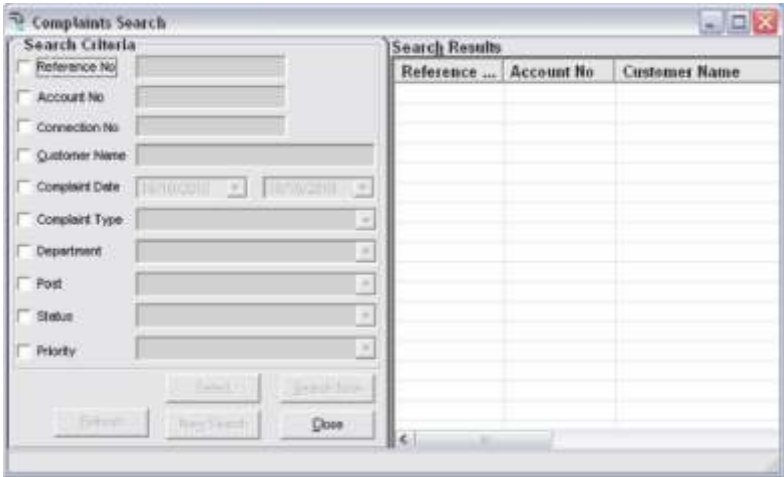


Figure 202: The Complaint Search Engine Screen

The reference number is what is needed to obtain the details of the complaint.
The user can use the account number of the customer, the connection number, names, date when the complaint was registered, type of complaint, department posted, post of the employee, status and priority.

12.4. PAYMENT SEARCH ENGINE

This function is found in the revenue section in the cancellation screen. It is used to obtain the payments reflected in the system made by a customer.



Figure 203: The Payment Search Engine Screen

It contains search criteria such as the account number of the customer, system receipt given out, old connection no, first name of the client, payment number, book receipt number, cheque number if payment was made via cheque, assigned bill, the payment date, the amount paid, the category and transaction type.

Click the search now button to initiate the search process once you have selected and entered the search criterion.

12.5. BILL SEARCH ENGINE

This function is used in the system to obtain bills that have been created previously and stored in the system. It can be found in the billing menu specifically in the Display Bill, Bill adjustment, Bill cancellation sub-menu when you want to show details of a bill.



Figure 204: The Bill Search Engine Screen

It includes search criteria with customer ID, Contract number, account Number, connection Number, First Name, Other Names, Bill Number, Due date, amount, category, billing group and billing period.

When you provide the available data and click search now button the closest result to your search string will be retrieved and displayed on the right hand side.

Select the correct one and click the select button to load the acquired bill to the form. If the bill is a periodical bill, then you have to select the billing period since by default the bill for the current billing month is what the search provides.

12.6. METER SEARCH ENGINE

This function is used in the system to obtain meters that can be found either on the client's site or the company's warehouse/store. It can be found in the meter management menu.

The following is the layout:



Figure 205: Meter Search Engine Screen

It contains meter search criteria such as the meter number, meter type, meter location, meter status and event type. You can also use the date search criteria which includes the last installation date (with an option of selecting the actual start and end date or a number of days from the present date).

When you provide available data in the search criteria and click the search now button the closest meter result to your search string will be retrieved and displayed at the bottom in the search results section.

Select the correct meter in question and click the select button to load the acquired meter to the form.

12.7. PARAMETER SEARCH ENGINE

This is a function that is used in the system at some search buttons to obtain constraints of some nature.

You can find this in Meter Reader selection, Meter Disconnectors, Meter Reconnectors and Modify Tariff. It can be viewed when you click the search button.

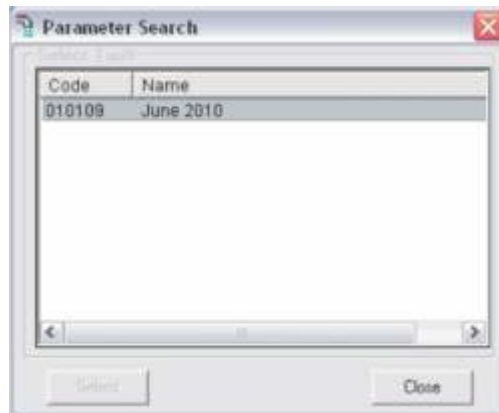


Figure 206: Parameter Search Selector Screen

Above is an example of a parameter search screen that is used to pick a tariff. When many options are available, once can select the desired one and click select to pick that option. This procedure works for the rest of the other modules too i.e. meter reader, disconnectors and reconnectors.